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Power Modeling for HSA Platform

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Outline

- Why the topic is important and difficult?
- Basics of power modeling of processors
- Challenges for power modeling of HSA

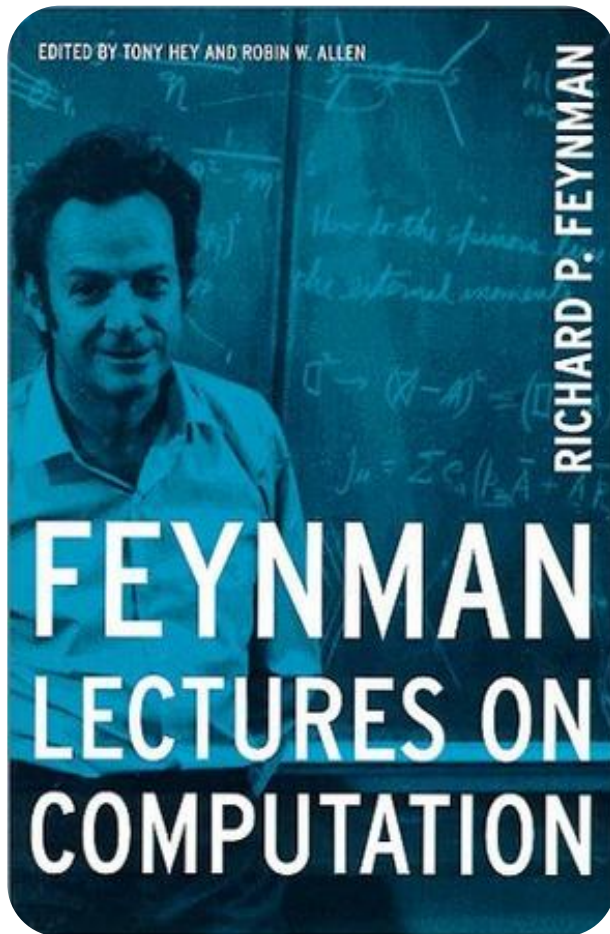


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Why the topic is important and difficult?

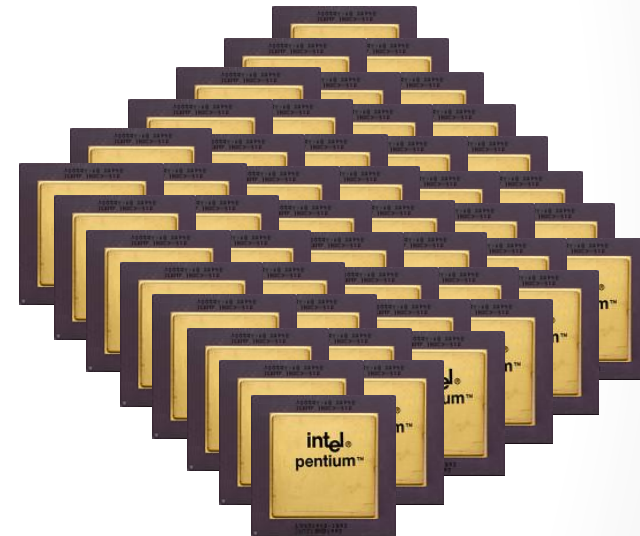
Why the topic is important?

- Energy inefficiency for computer systems
- How about battery operated devices?
 - Power bank
 - Wireless charging
- How about thermal issue?
 - Temperature bounded



COMPUTATION
RECKONED ON

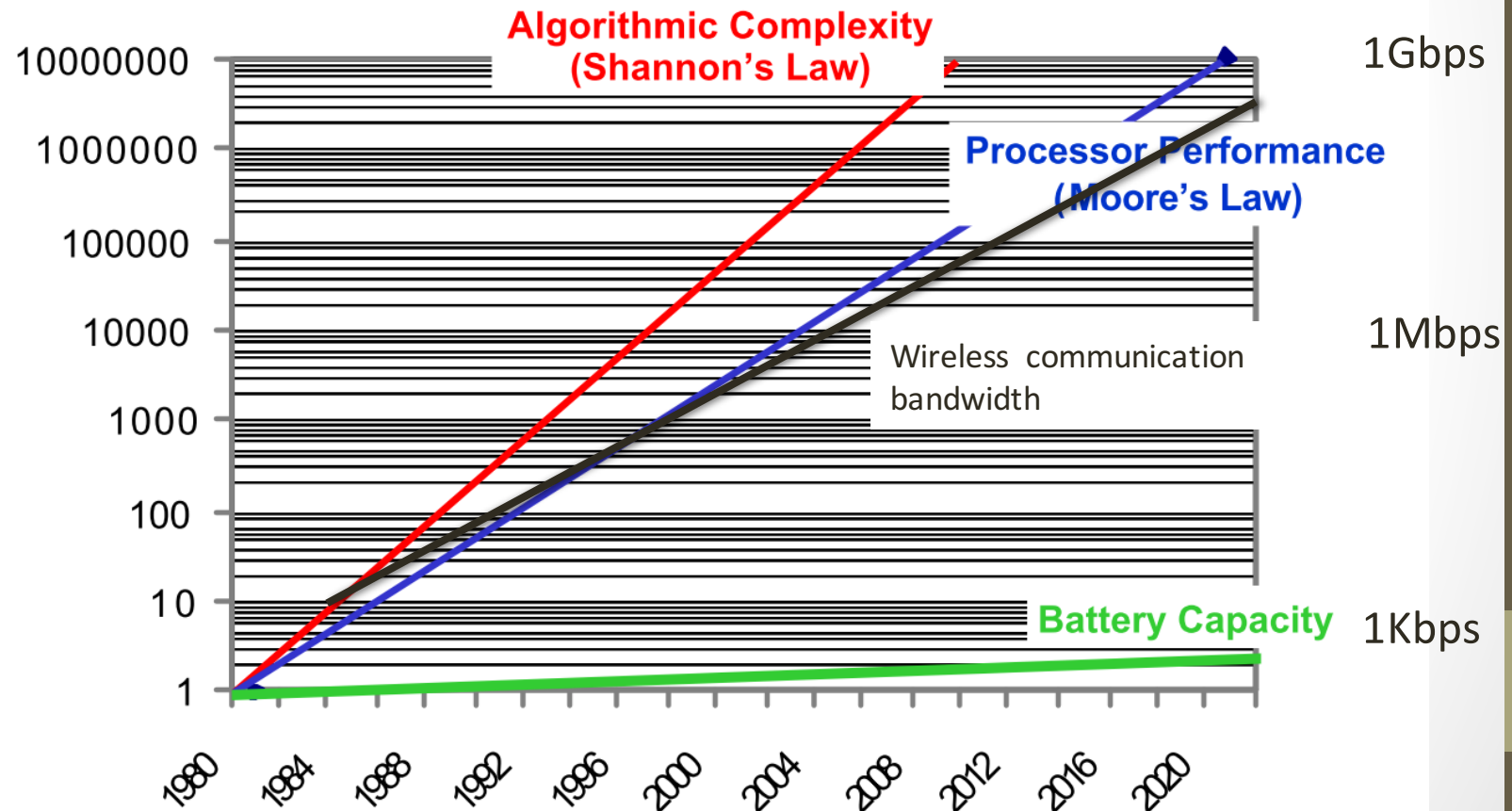
1 Watt for 10^{18} bit-ops/sec



About 1 billion Pentiums

Battery gap

- Computation and wired/wireless communication follow Moore's law



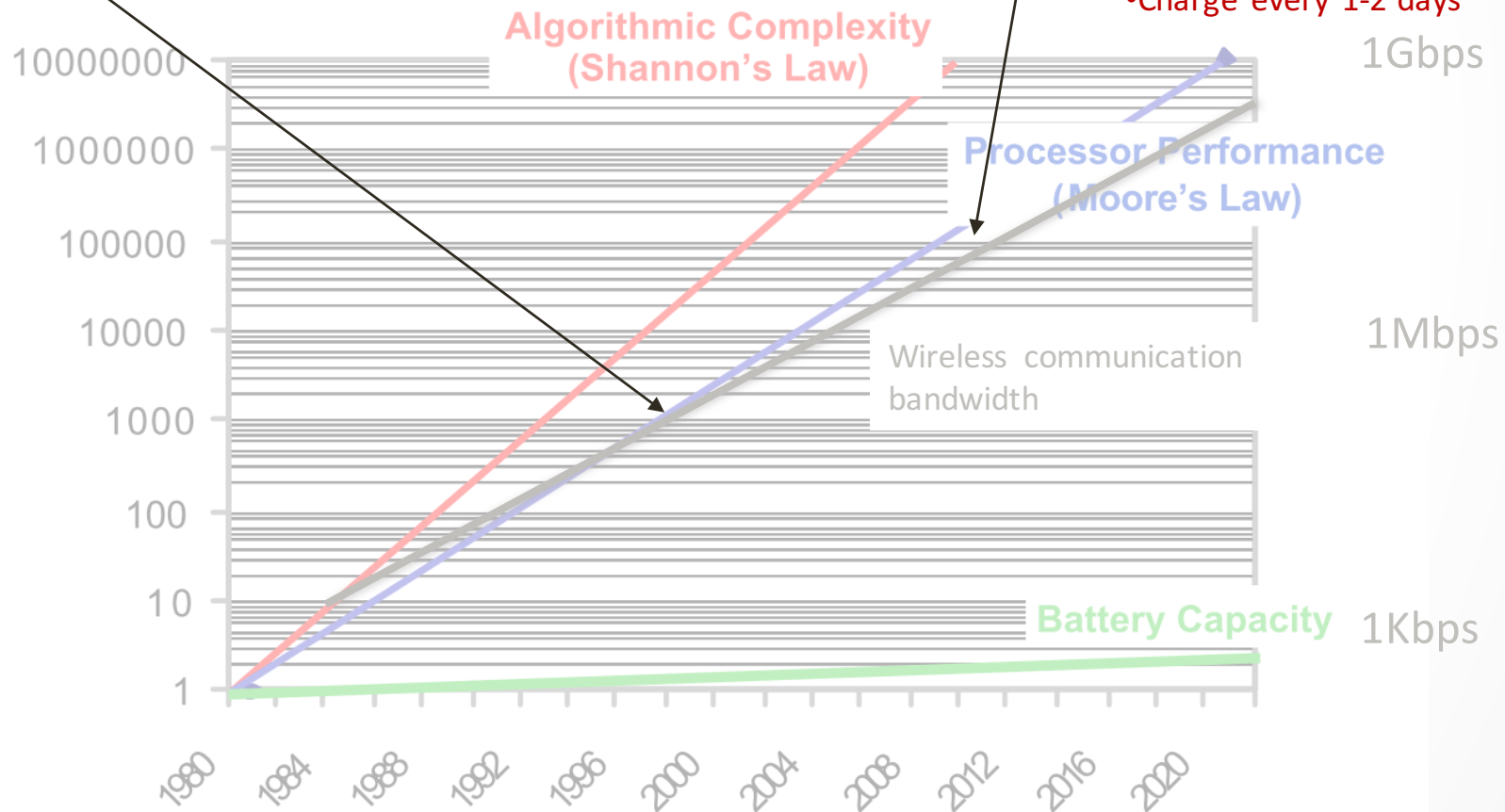
Is the problem that serious?



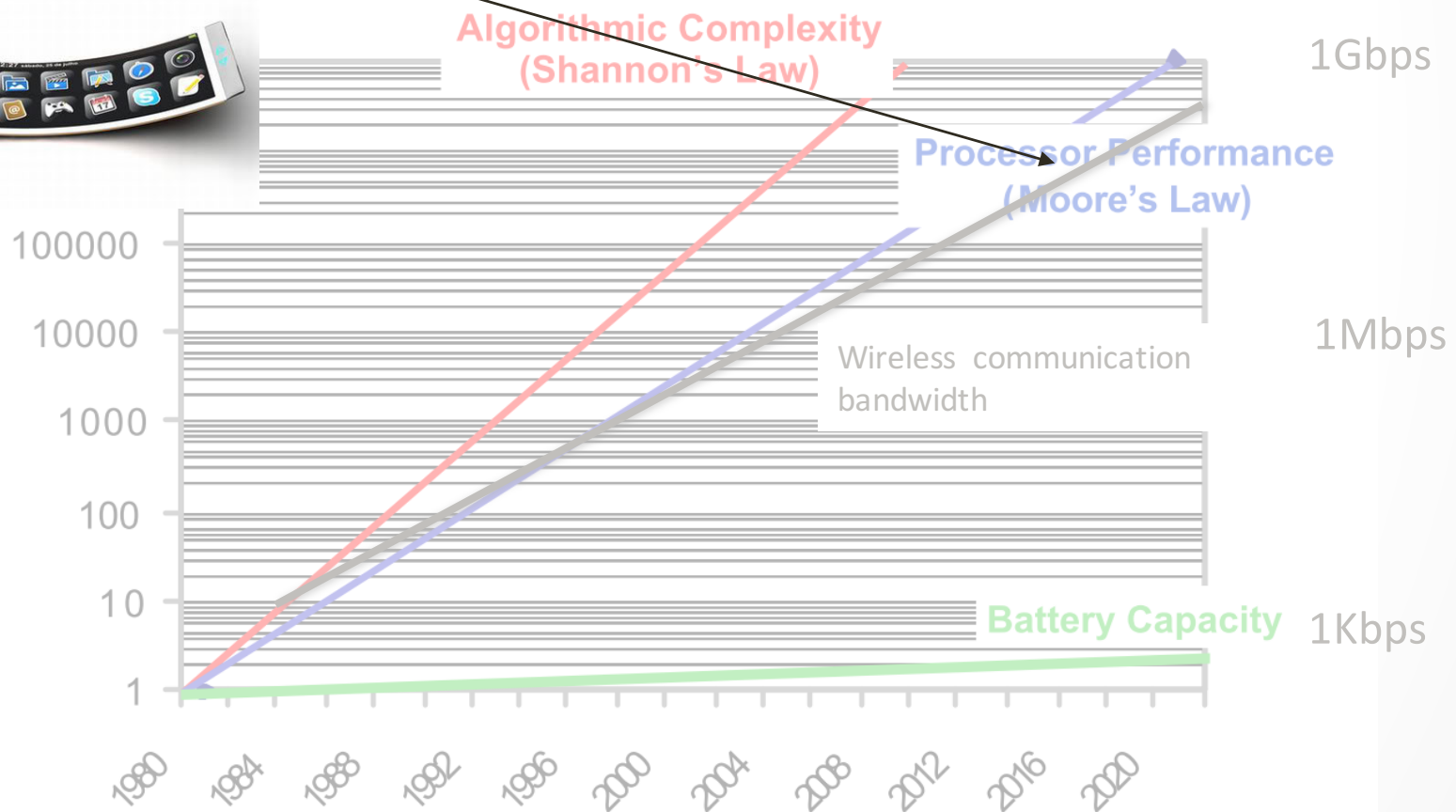
- ARM7TDMI@40MHz
- 128K SRAM
- 151 g
- BML-3 1200mAh
- GSM (@13kbps)
- Charge once a week



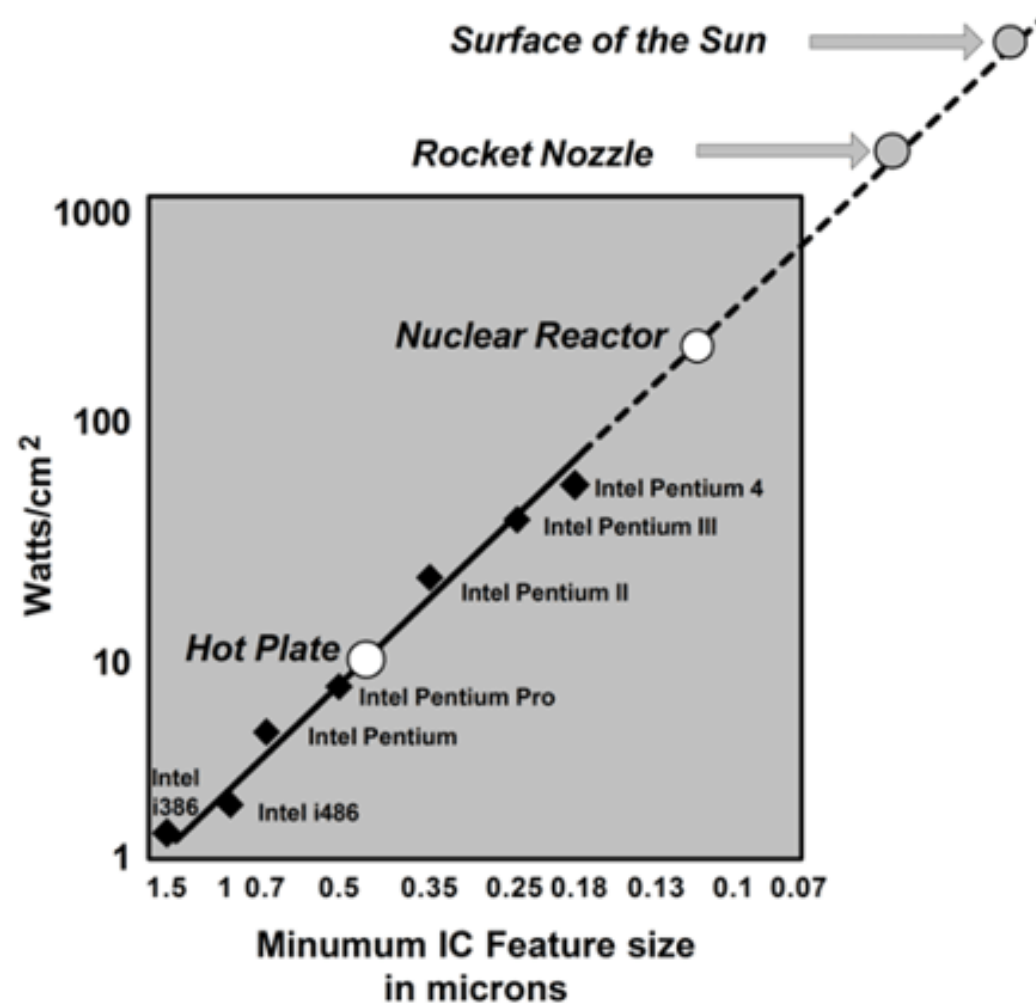
- Cortex-A8@1GHz GPU + PowerVR SGX535
- 512 MB RAM
- 137 g
- Li-Po 1420mAh
- HSDPA, 7.2 Mbps; HSUPA, 5.76 Mbps; 802.11b/g/n
- Charge every 1-2 days



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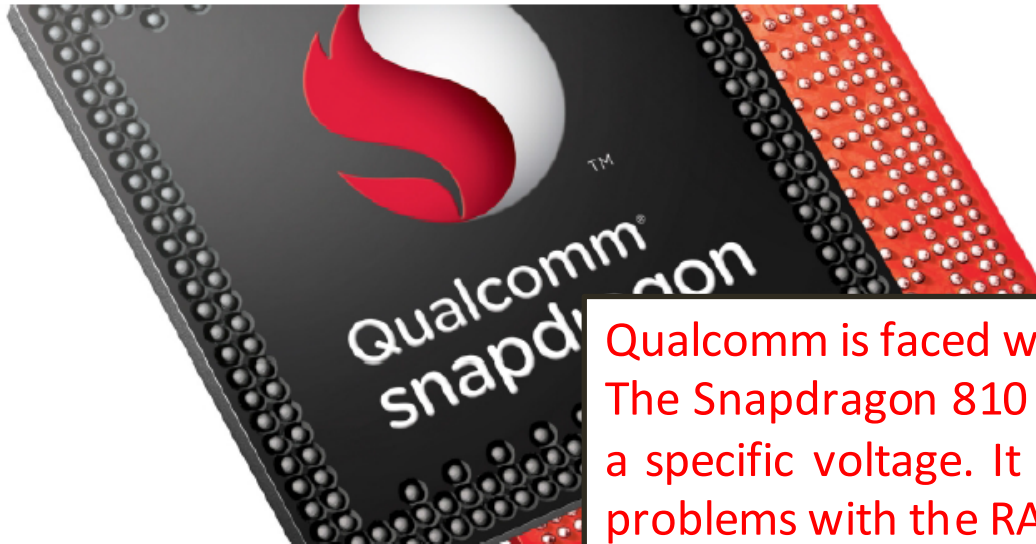


Power Density



Unexpected Hurdle

Problems in Qualcomm Snapdragon Set Alarm Bells Ringing for Samsung, LG



Qualcomm is faced with hard-to-solve problems. The Snapdragon 810 overheats when it reaches a specific voltage. It also slows down owing to problems with the RAM controller connected to the AP

4 DECEMBER 2014 | Cho Jin-young

Local smartphone makers are nervous at the prospect of a delay in the launch of new models next year, including the Galaxy S6 and the G4. It is unclear whether or not the supply of the Snapdragon 810 will exist in the first half of next year due to technical problems such as overheating and a decline in speed.

Samsung is likely to solve the problem by featuring its own Exynos chips in the Galaxy S6, but LG seems to be in trouble. Even though the company has its first AP, NUCLUN, it is not better than entry-level APs. If Qualcomm cannot supply the Snapdragon 810, it won't be easy for LG to find an alternative for the G4.

"Qualcomm is faced with hard-to-solve problems. The Snapdragon 810 overheats when it reaches a specific voltage. It also slows down owing to problems with the RAM controller connected to the AP. In addition, there is an error in the driver of the Adreno 430 GPU," said an industry source on Dec. 2.

The source added that for these reasons, it is unclear if the Snapdragon 810 will be used in premium smartphones like the Galaxy S6, the G4, and the Xperia Z4 scheduled to be released in the first half of next year.

The Snapdragon 810 is an octa-core AP composed of four quad-core A57s and four quad-core A53s, which supports 64-bit and an Ultra HD display.

Handset makers like Samsung and LG, which focus on premium smartphones, are likely to use technology that supports a UHD display and 64 bits in its next models. Hence, these companies badly need the Snapdragon 810.

Why the topic is difficult?

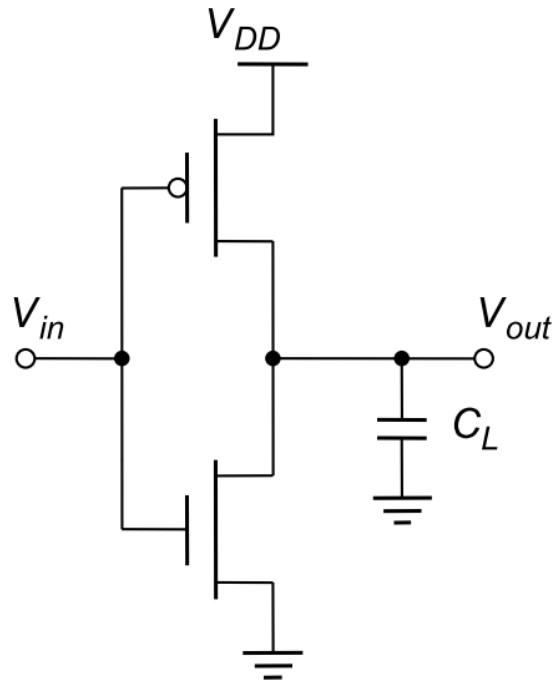
- Interdisciplinary issue
 - Thermodynamics, fluid mechanics
- Performance vs. power vs. thermal
 - Detection/action/reaction latency issues
- Power and thermal interaction issues



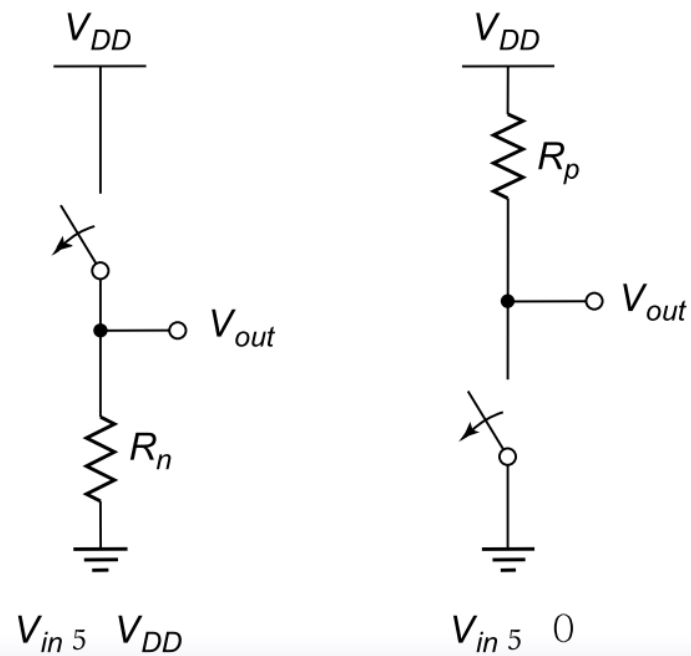
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Basics of power modeling of processors

The CMOS Inverter: A First Glance

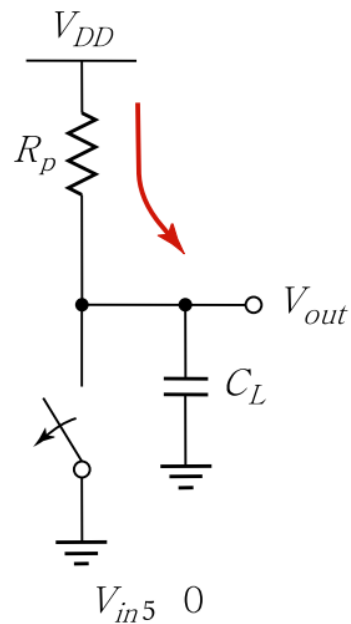


CMOS Inverter First-Order DC Analysis

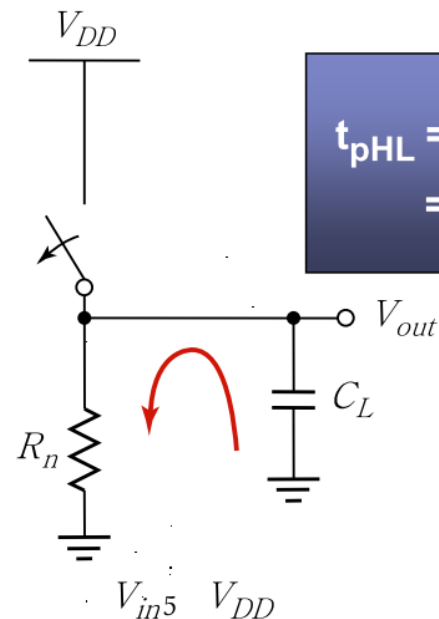


$$\begin{aligned} V_{OL} &= 0 \\ V_{OH} &= V_{DD} \\ V_M &= f(R_n, R_p) \end{aligned}$$

CMOS Inverter: Transient Response



(a) Low-to-high



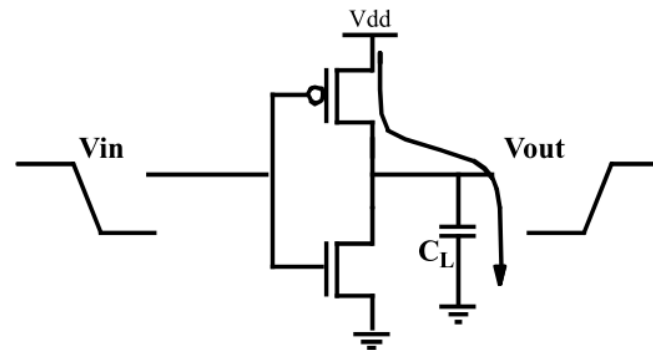
(b) High-to-low

$$t_{pHL} = f(R_{on} \cdot C_L) \\ = 0.69 R_{on} C_L$$

CMOS Power Consumption

- Dynamic Power Consumption
 - Charging and discharging capacitors
- Short Circuit Currents
 - Short circuit path between supply rails during switching
- Leakage
 - Leaking diodes and transistors

Dynamic Power Dissipation



$$\text{Energy/transition} = C_L * V_{dd}^2$$

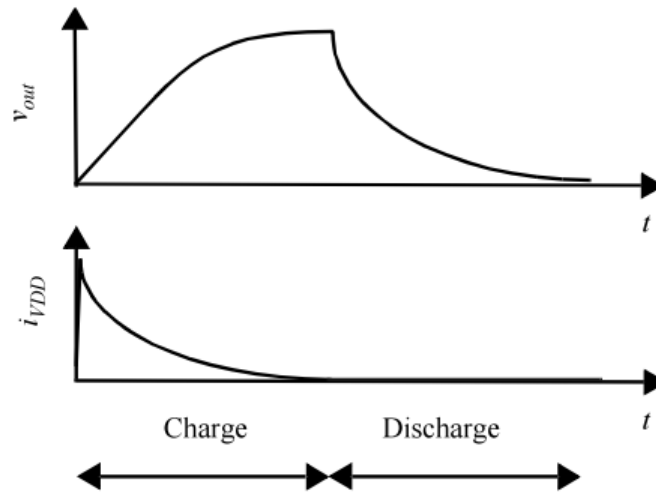
$$\text{Power} = \text{Energy/transition} * f = C_L * V_{dd}^2 * f$$

- Not a function of transistor sizes!
- Need to reduce C_L , V_{dd} , and f to reduce power.

$$E_{VDD} = \int_0^{\infty} i_{VDD}(t) V_{DD} dt = V_{DD} \int_0^{\infty} C_L \frac{dv_{out}}{dt} dt = C_L V_{DD} \int_0^{V_{DD}} dv_{out} = C_L V_{DD}^2$$

Source: Jan M. Rabaey, Anantha Chandrakasan, and Borivoje Nikolic, "Digital Integrated Circuits," A Prentice-Hall

Output voltages and supply current of capacitors



Source: Jan M. Rabaey, Anantha Chandrakasan, and Borivoje Nikolic, "Digital Integrated Circuits," A Prentice-Hall

Node Transition Activity and Power

- Consider switching a CMOS gate for N clock cycles

$$E_N = C_L \cdot V_{dd}^2 \cdot n(N)$$

E_N : the energy consumed for N clock cycles

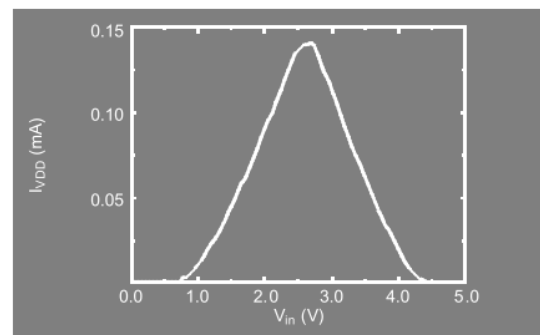
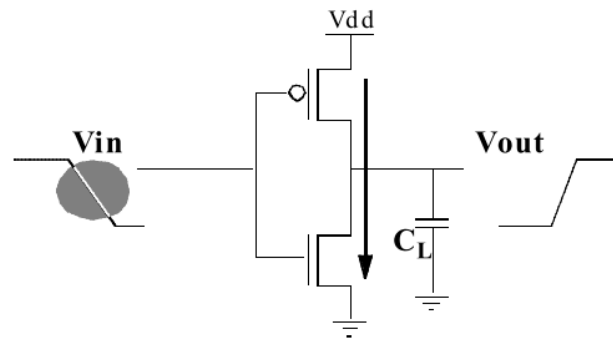
$n(N)$: the number of 0→1 transition in N clock cycles

$$P_{avg} = \lim_{N \rightarrow \infty} \frac{E_N}{N} \cdot f_{clk} = \left(\lim_{N \rightarrow \infty} \frac{n(N)}{N} \right) \cdot C_L \cdot V_{dd}^2 \cdot f_{clk}$$

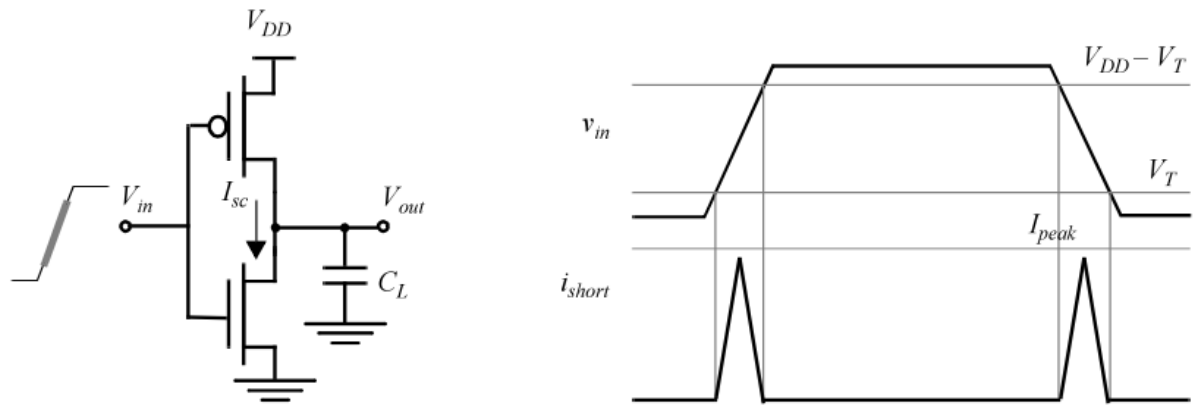
$$\alpha_{0 \rightarrow 1} = \lim_{N \rightarrow \infty} \frac{n(N)}{N}$$

$$P_{avg} = \alpha_{0 \rightarrow 1} \cdot C_L \cdot V_{dd}^2 \cdot f_{clk}$$

Short Circuit Currents



Short Circuit Power Consumption



$$E_{sc} = V_{DD} \frac{I_{peak} t_{sc}}{2} + V_{DD} \frac{I_{peak} t_{sc}}{2} = t_{sc} V_{DD} I_{peak}$$

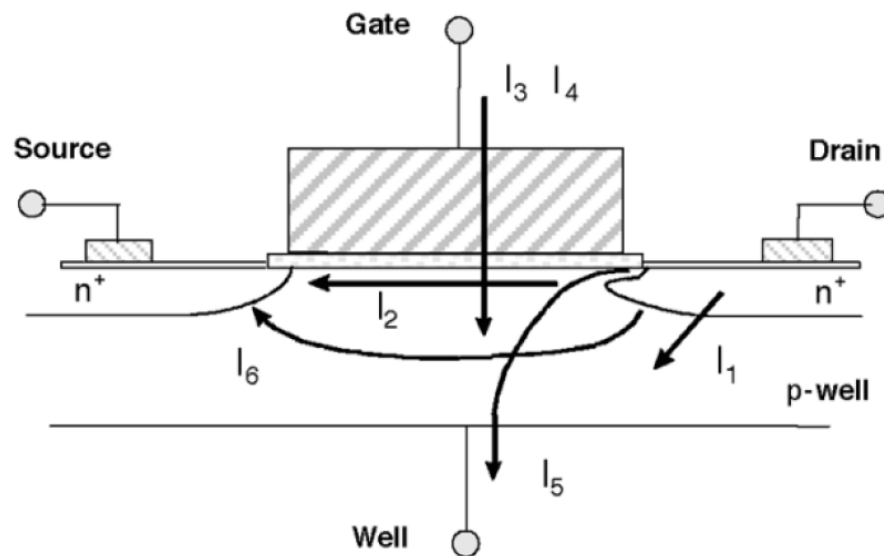
$$P_{sc} = t_{sc} V_{DD} I_{peak} f = C_{sc} V_{DD}^2 f$$

$$\frac{P_{sc}}{P_{dyn}} \approx 10\%$$

Source: Jan M. Rabaey, Anantha Chandrakasan, and Borivoje Nikolic, "Digital Integrated Circuits," A Prentice-Hall

Leakage Power Consumption

- There is no perfect insulator, there are always some leakage currents once IC is applied with a voltage



Source: K. Roy et al., *Leakage current mechanisms and leakage reduction techniques in deep-submicrometer CMOS circuits*, *Proceedings of the IEEE*, 2003

Leakage Power Consumption

- Gate leakage
 - tunneling current through the gate oxide insulation
- Subthreshold leakage
 - weak inversion current across the device
- Other leakages
 - but not as important as the above two items

Leakage Power Consumption

$$P_{stat} = I_{stat} V_{DD}$$

$$I_{stat} = I_{ox} + I_{sub}$$

$$I_{ox} = K_2 W \left(\frac{V}{T_{ox}} \right) e^{-\alpha T_{ox} / V}$$

$$I_{sub} = K_1 W e^{-V_{th} / n V_{\theta}} (1 - e^{-V / V_{\theta}})$$

Please refer to A. Chandrakasan, W. Bowhill, and F. Fox, "Design of High-Performance Microprocessor Circuits", IEEE Press, 2001. for detailed leakage power model

Total Power Consumption

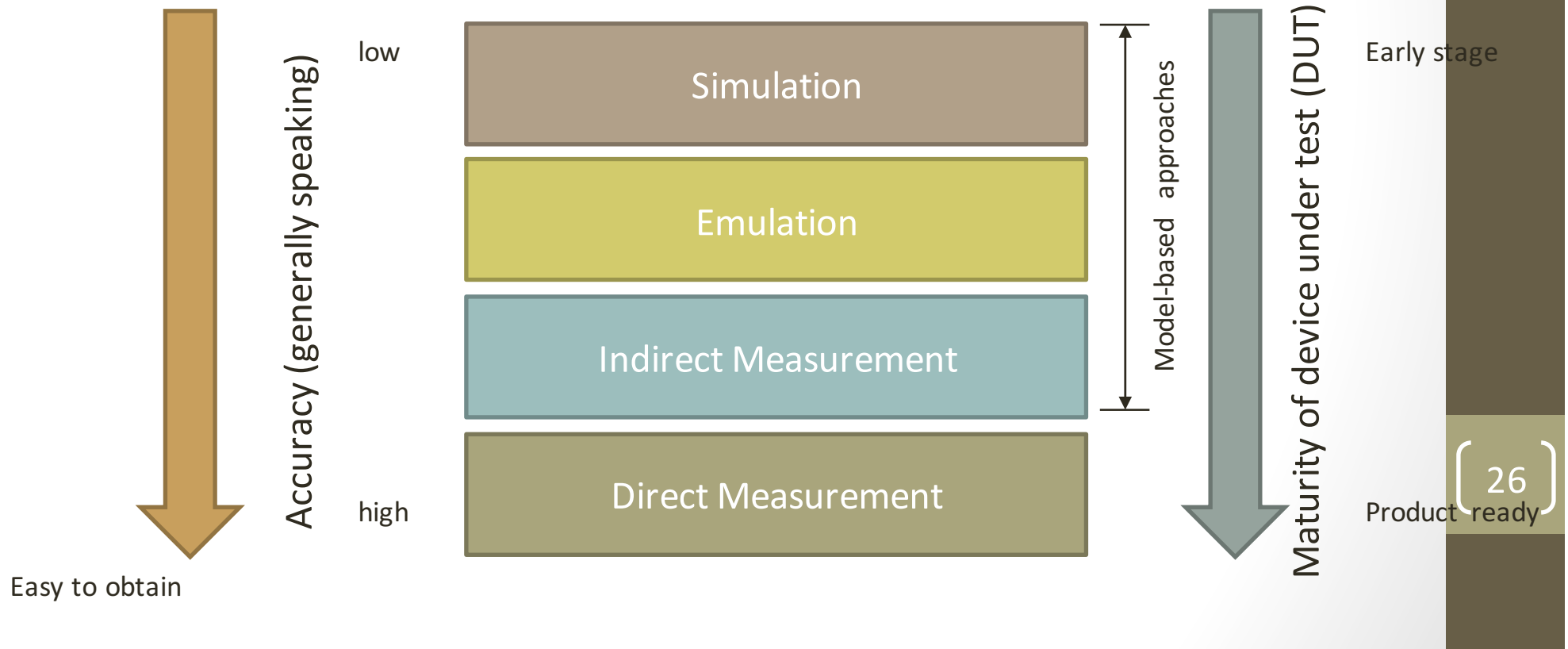
- CMOS power consumption model

$$\begin{aligned}P_{total} &= P_{dyn} + P_{sc} + P_{stat} \\&= (C_L V_{DD}^2 + V_{DD} I_{peak} t_s) f_{0 \rightarrow 1} + V_{DD} I_{stat} \\&= (C_L V_{DD}^2 + C_{sc} V_{DD}^2) f_{0 \rightarrow 1} + V_{DD} I_{stat}\end{aligned}$$

Overview

- “If you can not measure it, you can not improve it.” by Lord Kelvin

Difficult to obtain (both technologies and time)



Overview

- Direct measurement
 - Directly measure the current and voltage of a DUT or components of the DUT while DUT is running
 - Precise results for DUT and/or components of DUT
- Indirect measurement
 - Collect data such as total power, performance counters, etc. while DUT is running, and use the data to estimate power consumption of DUT and/or components of DUT
 - Energy models which map the data to power are required

Overview

- Emulation
 - similar to simulation but the DUT is running on a real platform such as FPGA prototype
 - Speedup the simulation time
- Simulation
 - To simulate power consumption through circuit, gate, instruction, architecture, etc. models

Overview

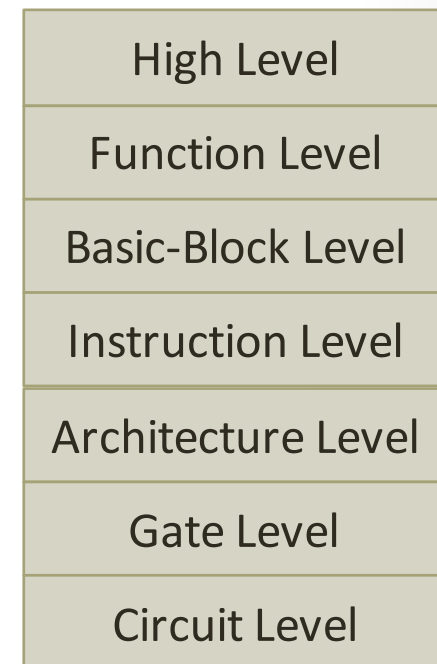
- Measurement-based approaches are relatively easy but they can be applied to the systems which are already there
- Precise measurements need expensive hardware
- Model-based approaches can be applied to the hardware/software design phase

Overview

- Issues with model-based approaches
 - It takes time to simulate all signals through circuits. How long does it take to simulate a program? Can we get the circuit design of embedded system?
 - Can we simplify it by estimating gate activities?
 - Can we simplify it by estimating hardware blocks activities?
 - Can we simplify it by estimating instruction execution?
 - Can we simplify it by estimating function/process execution?

Overview

- Issues with model-based approaches
 - How detail information of the design we can get
 - Circuit, RTL, instruction, functions, ...
 - How long we can wait for the results
 - How precise information we need



(31)

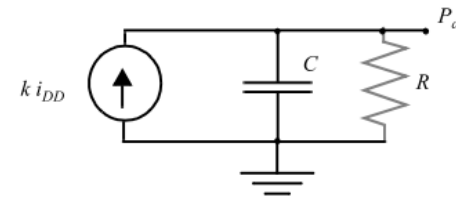
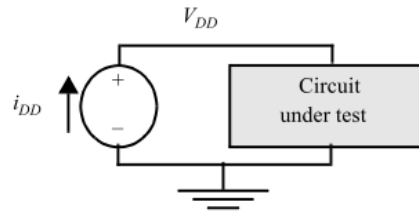
Circuit Level Simulation

- SPICE (Simulation Program with Integrated Circuit Emphasis) can measure the average value of a circuit signal

$$C \frac{dP_{av}}{dt} = k i_{DD}$$

or

$$P_{av} = \frac{k}{C} \int_0^T i_{DD} dt$$

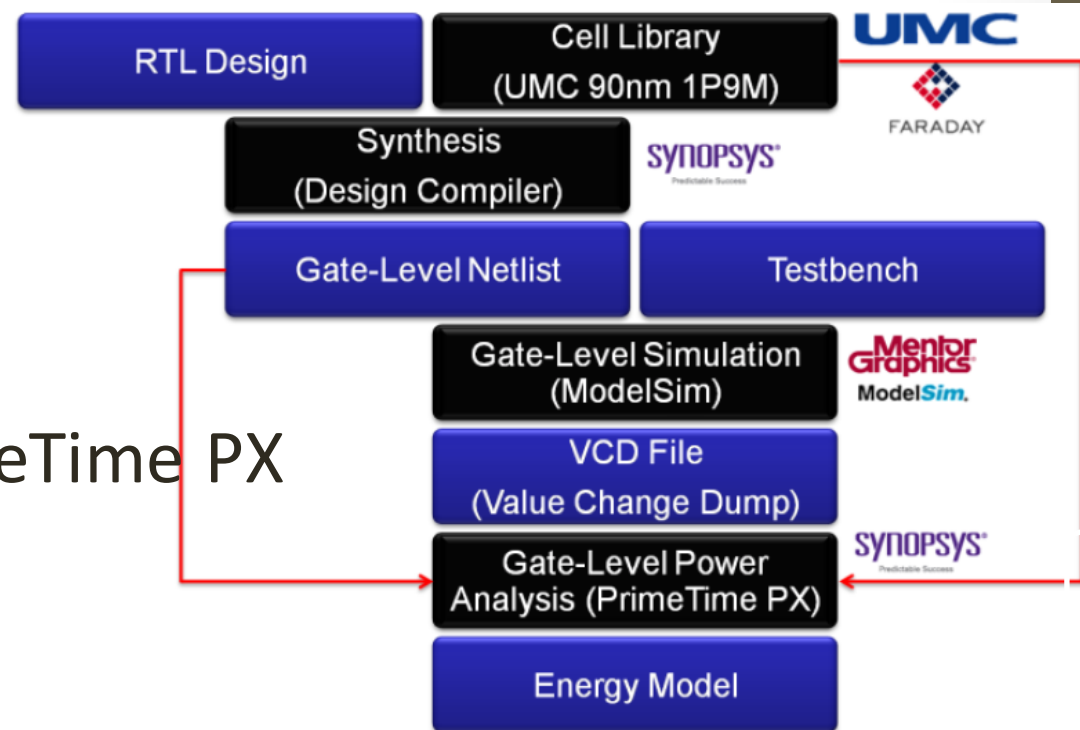


- e.g. Synopsys HSPICE

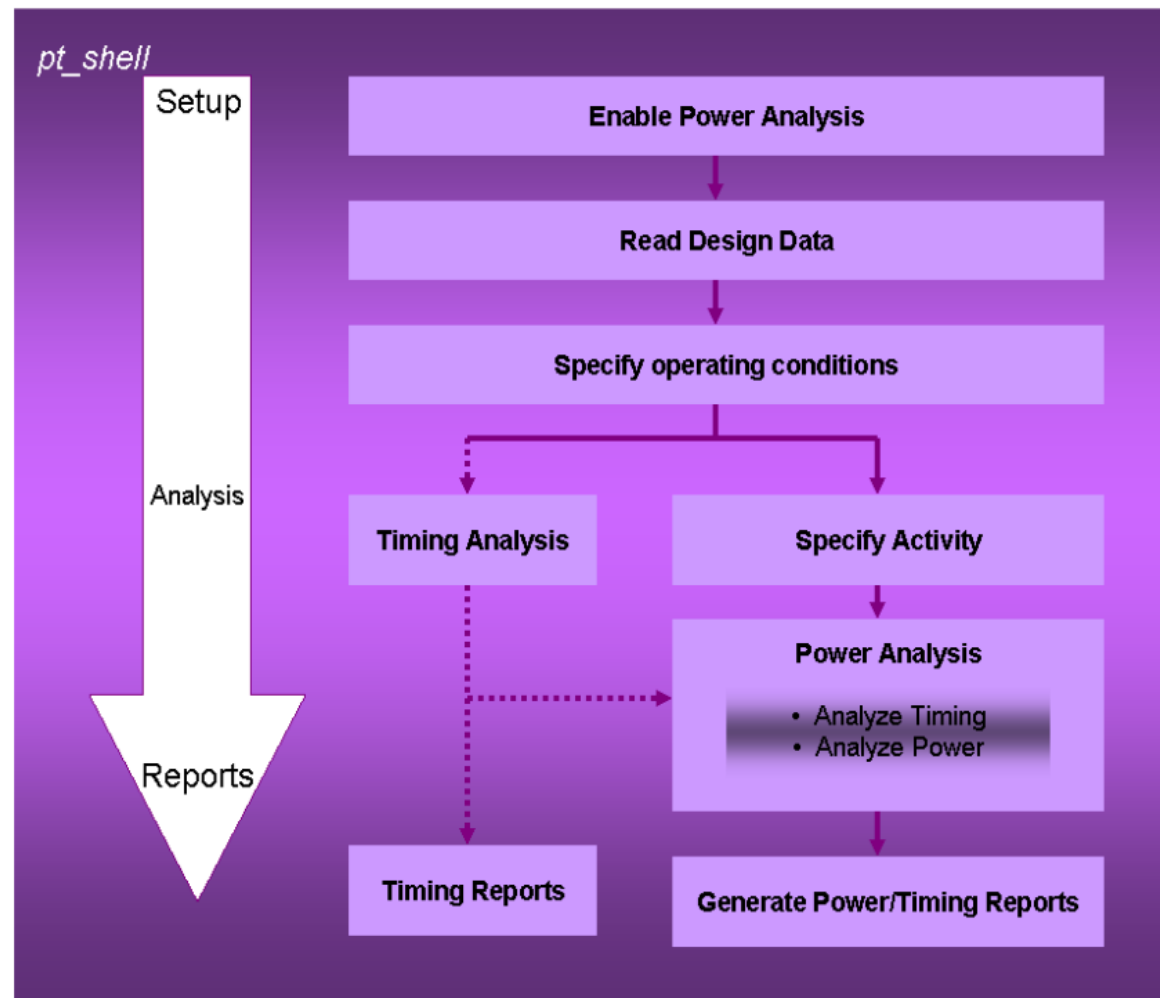
Gate Level Simulation

- Calculate power consumption of gates based on given CMOS production parameters and a predefined percentages of gate activities

- e.g. Synopsys PrimeTime PX



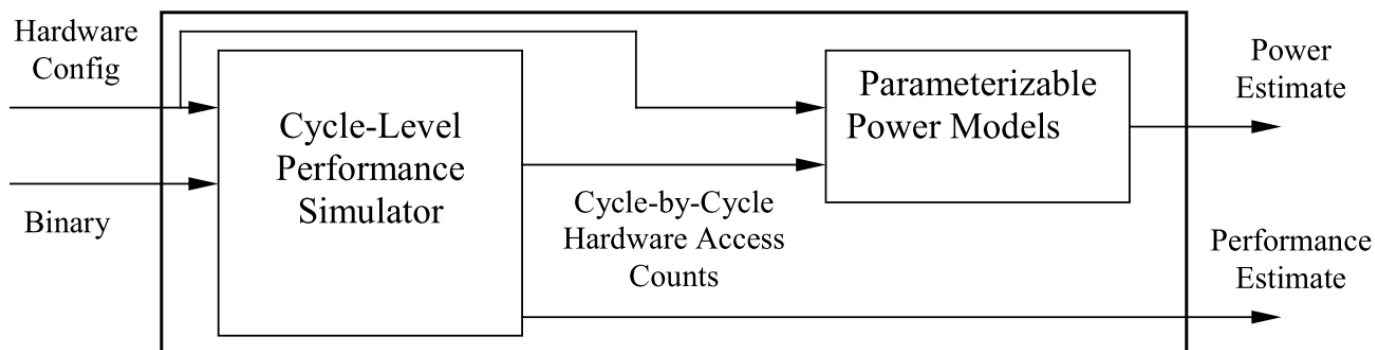
Gate Level Simulation



Architecture Level Simulation

- Parameterized power models for common structures in processors
- Run programs on architecture simulators (such as instruction set simulators)
- Collect activities on these structures from simulators
- Calculate the power consumption of the processor

Wattch



Hardware Structure	Model Type
Instruction Cache	Cache Array (2x bitlines)
Wakeup Logic	CAM
Issue Selection Logic	Complex combinational
Instruction window	Array/CAM
Branch Predictor	Cache Array (2x bitlines)
Register File	Array (1x bitlines)
Translation Lookaside Buffer	Array/CAM
Load/Store Queue	Array/CAM
Data Cache	Cache Array (2x bitlines)
Integer Functional Units	Complex combinational
FP Functional Units	Complex combinational
Global Clock	Clock

Four categories of processor units

$$P_d = CV_{dd}^2 af$$

Process technology

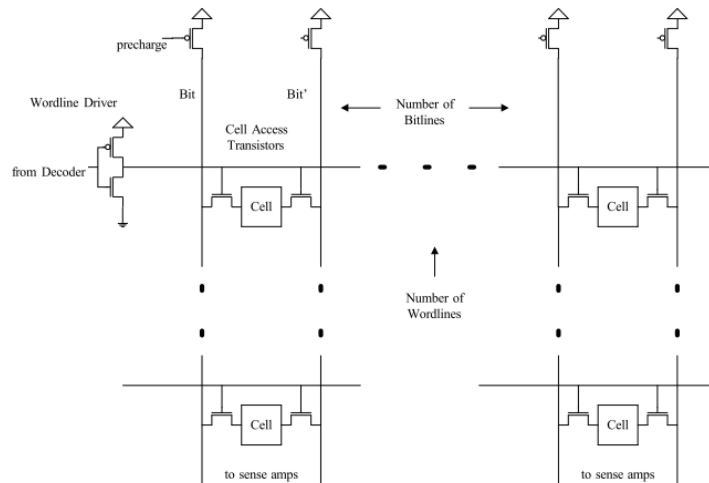
model

Collected from simulator or based on certain assumptions

- Array structures: data and instruction caches, register files, ...
- Fully associative content-addressable memories: TLBs, ...
- Combinational logic and wires: functional units, ...
- Clocking: clock buffers, clock wires, ...

Capacitances Model

- Array structure example



Node	Capacitance Equation
Regfile Wordline Capacitance =	$C_{diff}(WordLineDriver) + C_{gate}(CellAccess) * NumBitlines + C_{metal} * WordLineLength$
Regfile Bitline Capacitance =	$C_{diff}(PreCharge) + C_{diff}(CellAccess) * NumWdlines + C_{metal} * BLength$
CAM Tagline Capacitance =	$C_{gate}(CompareEn) * NumberTags + C_{diff}(CompareDriver) + C_{metal} * TLength$
CAM Matchline Capacitance =	$2 * C_{diff}(CompareEn) * TagSize + C_{diff}(MatchPreCharge) + C_{diff}(MatchOR) + C_{metal} * MLength$
ResultBus Capacitance =	$.5 * C_{metal} * NumALU * ALUHeight + C_{metal} * (RegfileHeight)$

Multi-core Performance Area and Timing McPAT

New evaluation metric (Operational cost and capital cost)

- EDAP or EDA²P (die cost $\propto A^2$)

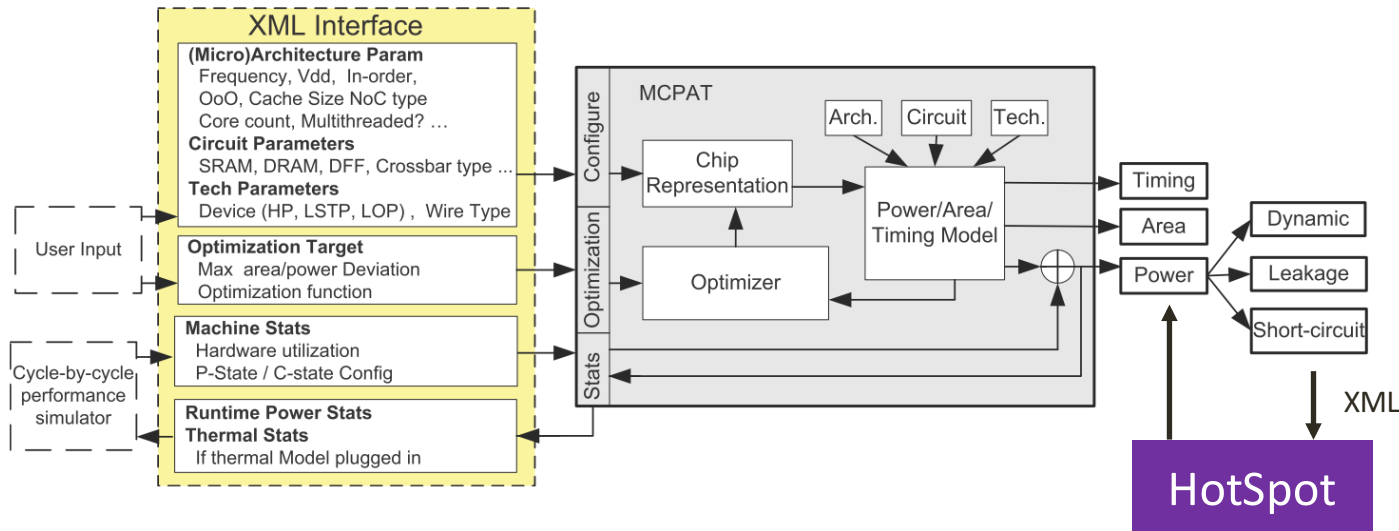
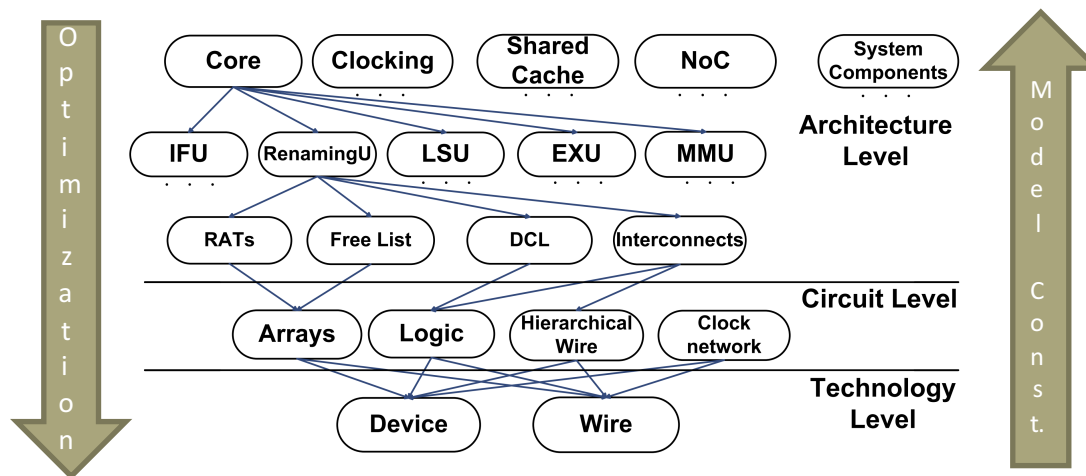


Figure 1: Block diagram of the McPAT framework.



McPAT Features

- Abstracting micro-architectural parameters of each components
- $P = \alpha CV^2 f + \alpha E_{short} f + I_{leakage} V$
- Timing model compute RC delay
- Area model uses
 - analytical equations:
 - empirical approach building parameterizable numeric model
- Clock and power domains

Processor	Published total power	Modeled total power	McPAT results	% McPAT error
Niagara	63W	61.4W	56.17W	-10.84
Niagara2	84W	77.9W	69.70W	-17.02
Alpha 21364	125W	119.8W	97.9W	-21.68
Xeon Tulsa	150W	145.5W	116.08W	-22.61

Table 3: Validation results of McPAT with regard to total power of target processors.

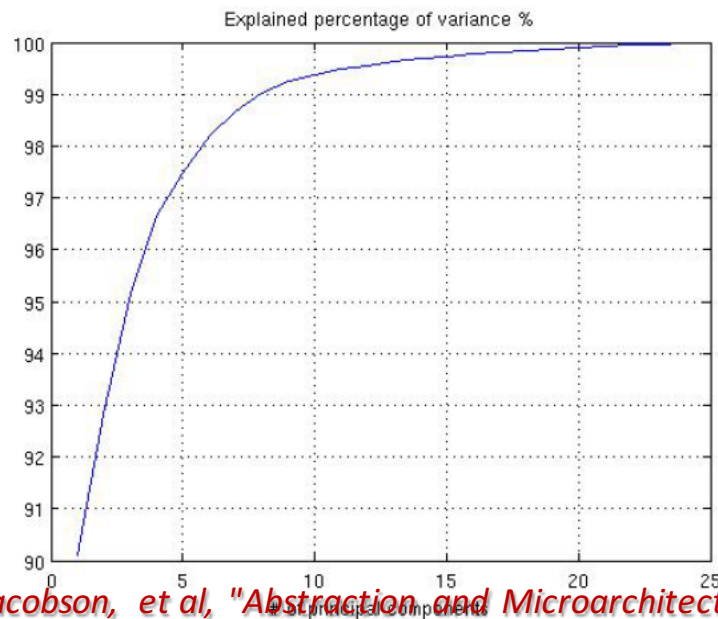
Processor	Published die size (mm^2)	McPAT results (mm^2)	McPAT error %
Niagara	378	295	-21.8
Niagara2	342	248	-27.3
Alpha 21364	396	324	-18.2
Xeon Tulsa	435	362	-16.7

Table 4: Validation Results of McPAT with regard to chip die area of target processors.

ROP, Video decode ckt.

Architecture Level Simulation

- How many parameters we should observe?
- Are they related to each other? (orthogonal)
- How important they are?



roughly 20-25 principal components are sufficient to explain the variability in 2300 attributes used

the first 8 principal components, we can explain more than 99% of the variability in the dataset (41)

Source: H. Jacobson, et al, "Abstraction and Microarchitecture Scaling in Early-Stage Power Modeling," in the 17th International Symposium on High Performance Computer Architecture (HPCA), Feb. 2011.

Instruction Level Simulation

- How if we cannot obtain the processor internal design?
 - Cannot derive capacitances and measure the activities
- Energy consumption is strongly related to the instructions that processor executes
- Assume processor is a block box, and analyze the instructions to derive the power consumption

Instruction Level Simulation

- Why instruction level simulation is inaccurate?
 - Power consumption of an instruction depends on sequences (instruction cache), data availability (data cache), operations bit patterns, ...
 - Other factors such as pipeline, branch prediction, cache design, superscalar, out-of-order execution, vector instructions influence the power consumption of a processor

Instruction Level Simulation

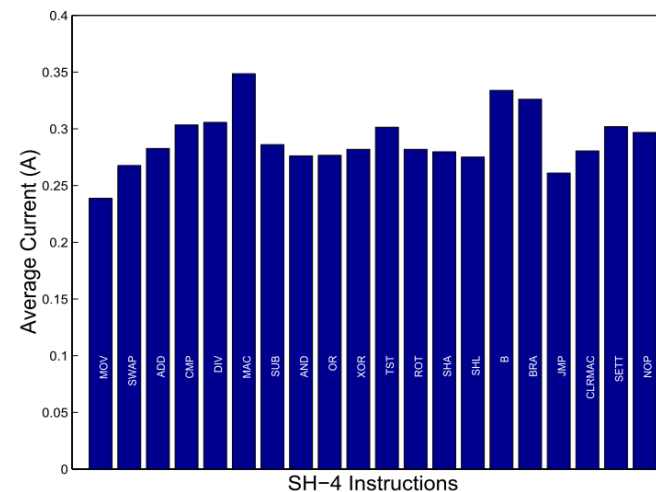
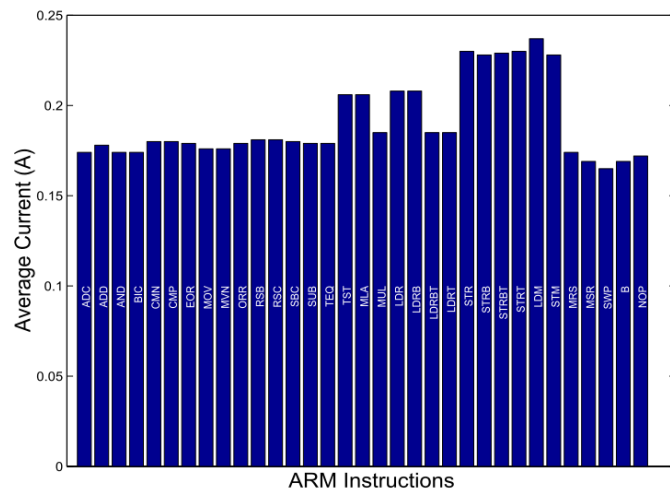
Number	Instruction	Current (mA)	Cycles
1	NOP	275.7	1
2	MOV DX,BX	302.4	1
3	MOV DX,[BX]	428.3	1
4	MOV DX,[BX][DI]	409.0	2
5	MOV [BX],DX	521.7	1
6	MOV [BX][DI],DX	451.7	2
7	ADD DX,BX	313.6	1
8	ADD DX,[BX]	400.1	2
9	ADD [BX],DX	415.7	3
10	SAL BX,1	300.8	3
11	SAL BX,CL	306.5	3
12	LEA DX,[BX]	364.4	1
13	LEA DX,[BX][DI]	345.2	2
14	JMP label	373.0	3
15	JZ label	375.7	3
16	JZ label	355.9	1
17	CMP BX,DX	298.2	1
18	CMP [BX],DX	388.0	2

BASE COSTS OF MOV BX, DATA

data	0	0F	OFF	OFFF	OFFFF
No. of 1's	0	4	8	12	16
Current(mA)	309.5	305.2	300.1	294.2	288.5

Instruction Level Simulation

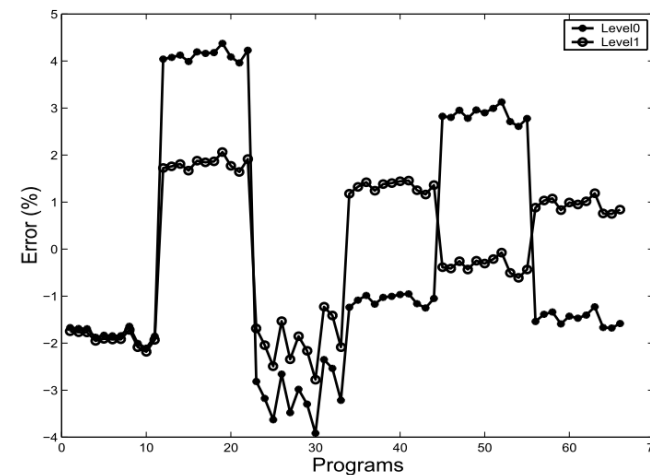
- RISC processor has a more consistent power consumption behaviors for the instructions



Instruction Level Simulation

- Categorize into four types of instructions

Class	Weight	Value
Instruction	w_1	2.1739
Sequential memory access	w_2	0.0311
Non-sequential memory access	w_3	1.2366
Internal cycles	w_4	0.8289



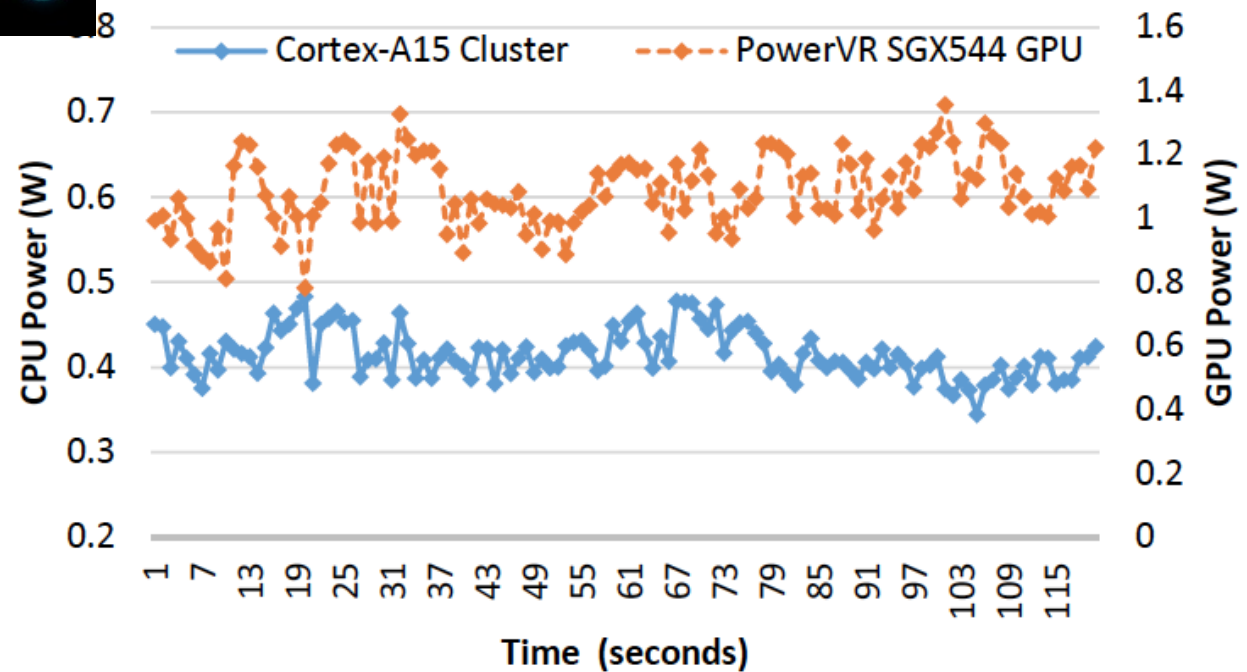
Basic-block Level Simulation

- Consider the power consumption of each basic block (without branch) rather than the power consumption of each instruction
- Gain speedup, lose certain accuracy

Function Level Simulation

- Assume that power consumptions of functions (no matter the input data and execution paths) are similar
- Consider the power consumption of each function rather than the power consumption of each basic block

Benchmark results



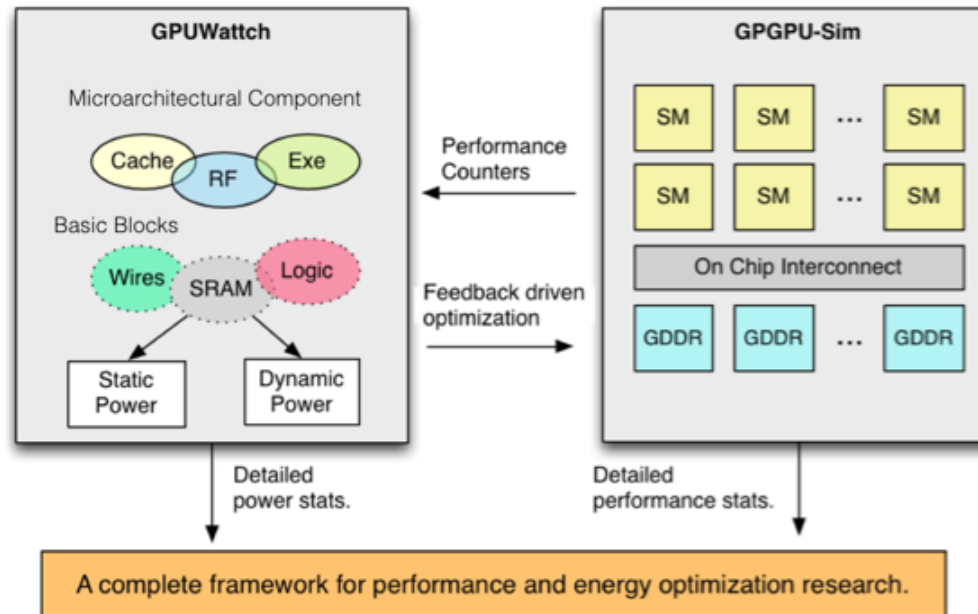
Targets

- Desktop GPU vs. Embedded GPU
 - Fast vs. energy efficiency
- GPU vs. GPGPU
 - Focus on graphic pipelines on generic GPU architecture
 - GPGPU + hardware accelerator

GPU Simulator

Simulators Features	GPGPU-Sim	Multi2Sim	ATTILA	TEAPOT
Open Source	Yes	Yes	Yes	No
GPGPU Support	CUDA and OpenCL	CUDA 4.0 and OpenCL 1.1	No	No
OpenGL Support	None	OpenGL 4.2	OpenGL 1.1 and DirectX10	OpenGL ES 2.0
Graphics Pipeline Implementation	None	None	IMR	TBIMR
Power Model	GPU Wattch / GPUSimPow	McPat	None	McPat
Last Release	2013	2013	2011	2013
First Targets	Nvidia GPU	AMD GPU	ATI / Nvidia GPU	Unknown
Purpose	GPGPU	Heterogeneous Computing	Graphics	Full Mobile GPU System

GPU Power Models



GPUWattch

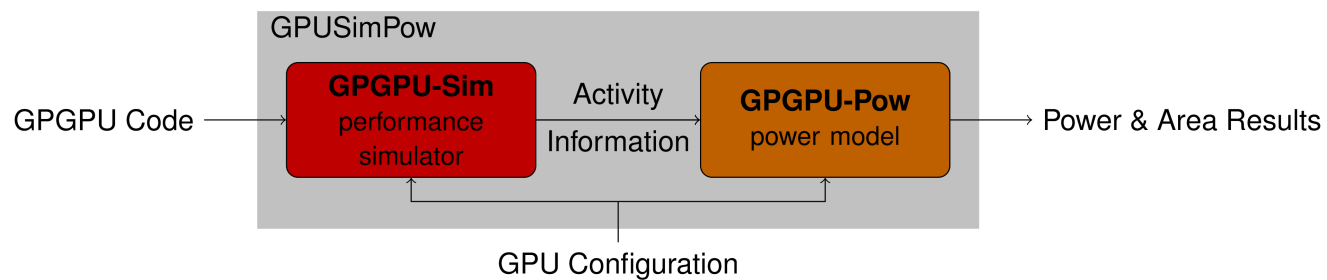


Fig. 1: Overview over the GPUSimPow simulation framework.

CPU and GPU Micro-Architecture

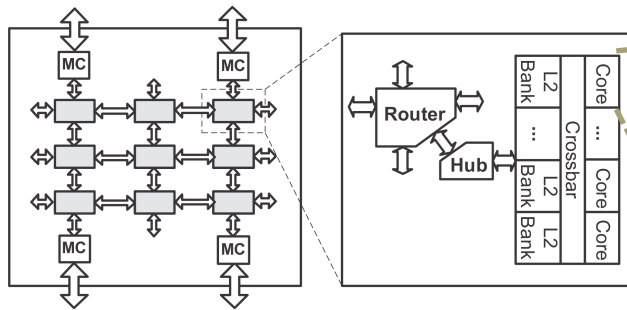
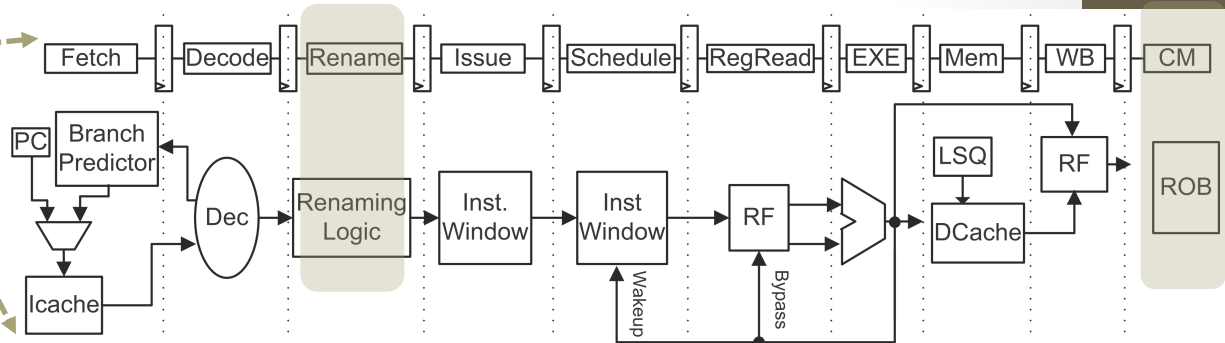
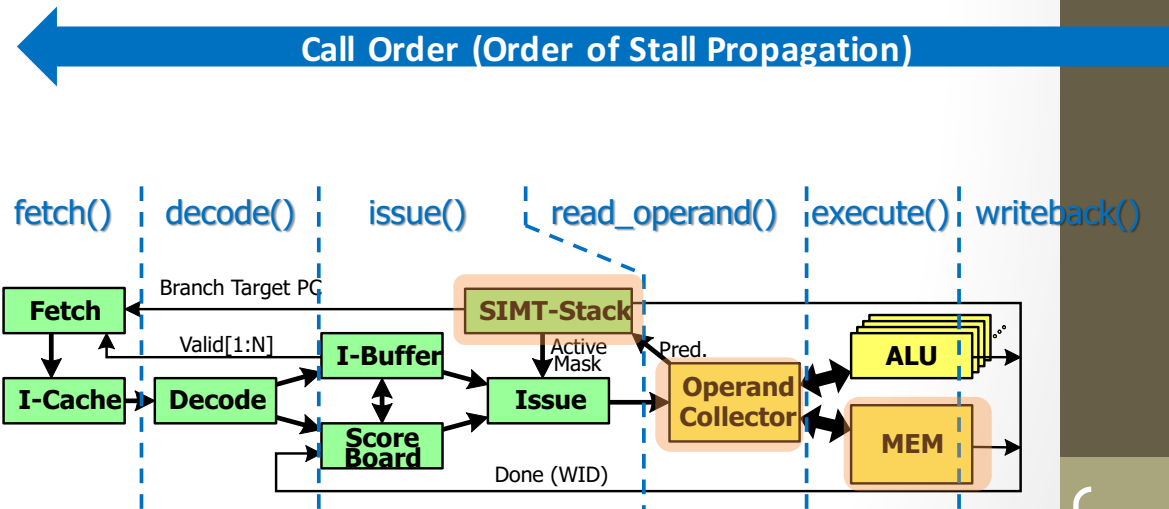
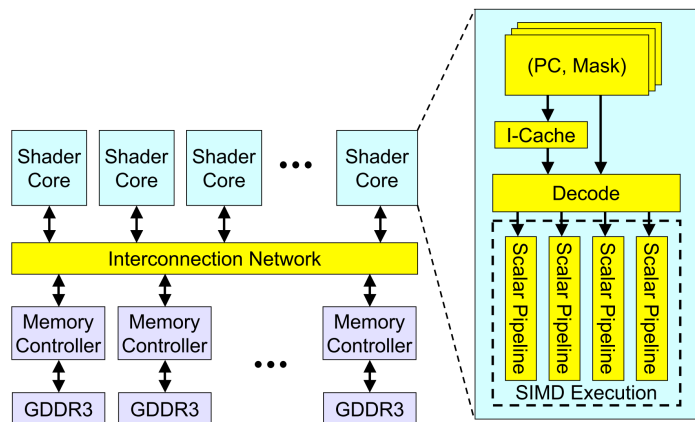


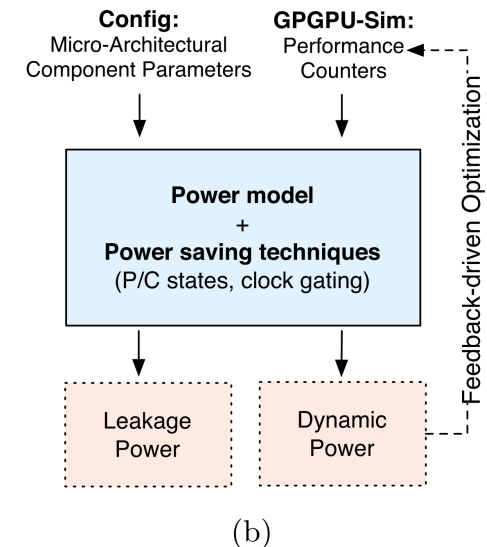
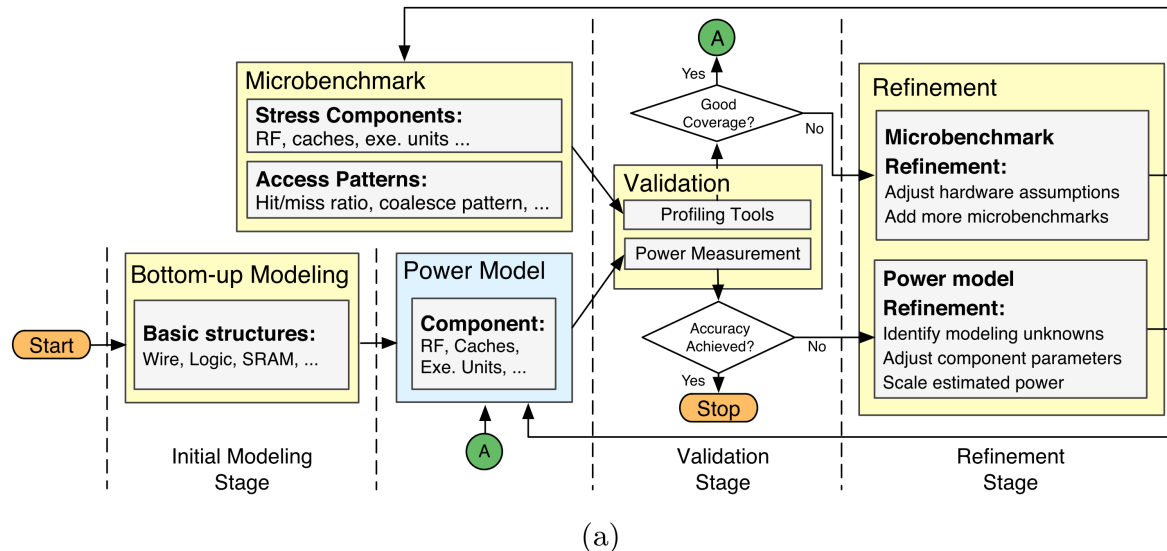
Figure 4: Manycore system architecture. MCs refer to memory controllers.



(a) McPAT model for physical register based OOO core.



GPUWattch



- Configurable
- Cycle level
- Validated (GTX 480 and Quadro FX5600)
 - 80 micro-benchmarks
 - 25 real-world kernels

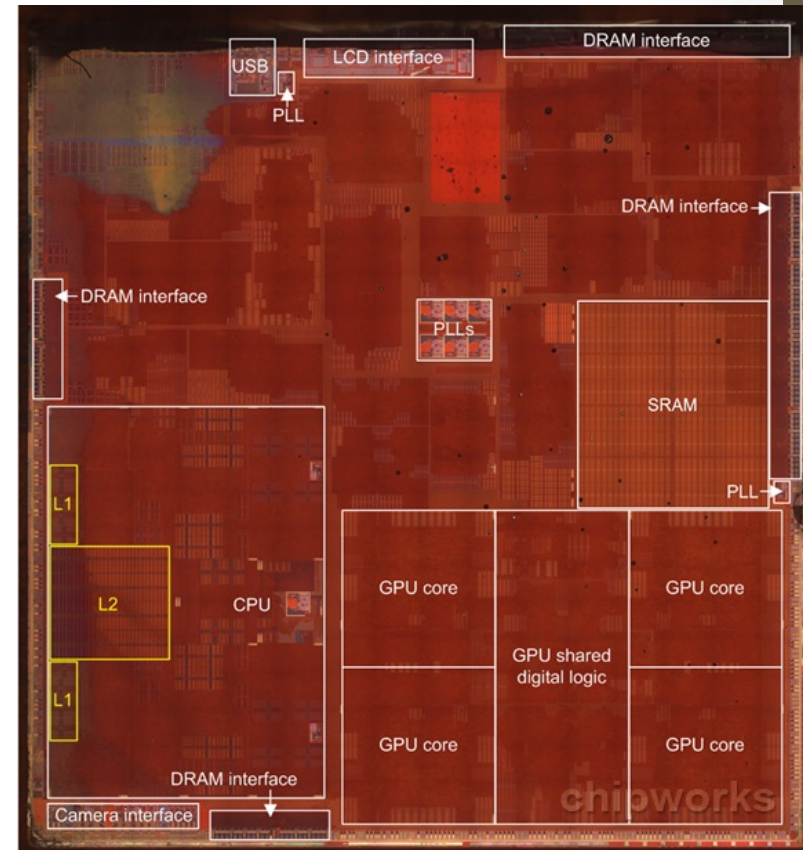


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Challenges for power modeling of HSA

Power Budget Issue

- Apple A7
 - GPU fills 22% of die
 - Dual-core 64-bit ARM
 - 1.1B transistors
 - 102 mm² @ 28nm CMOS

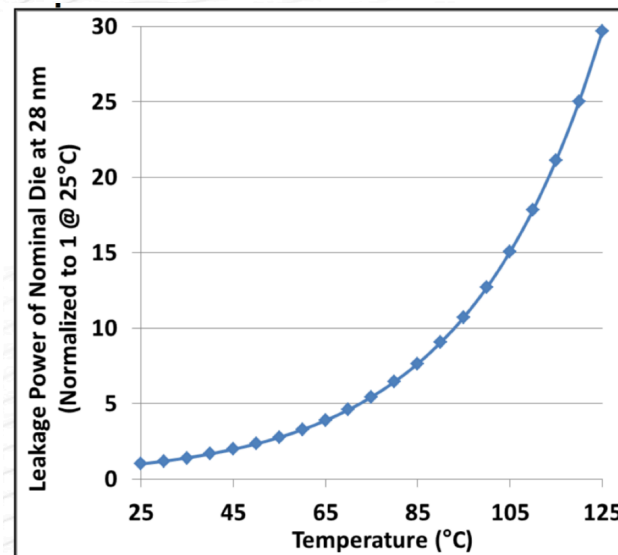
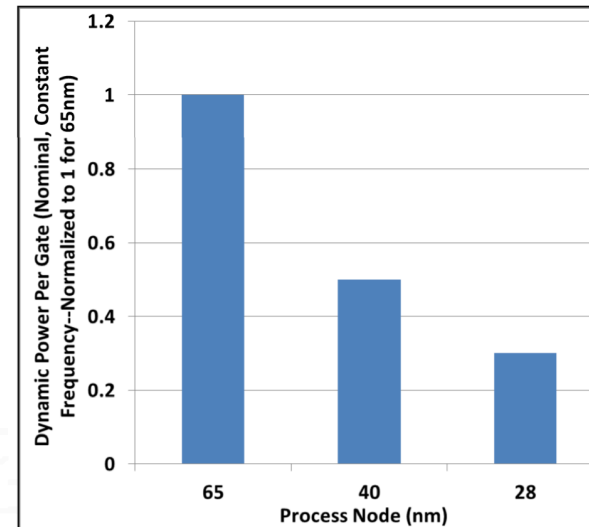
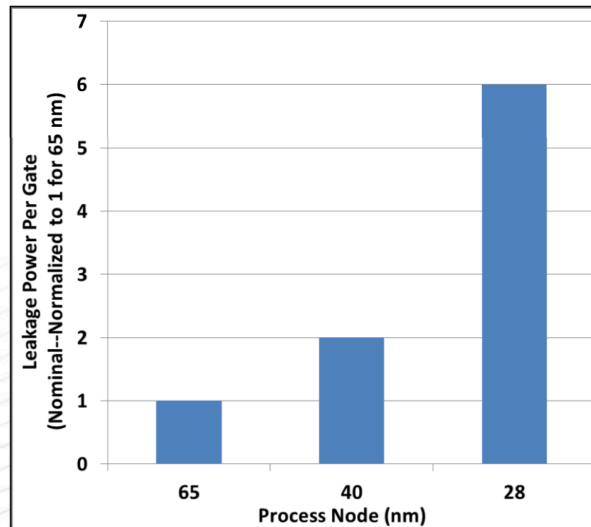


GVASS

green computing & embedded system lab

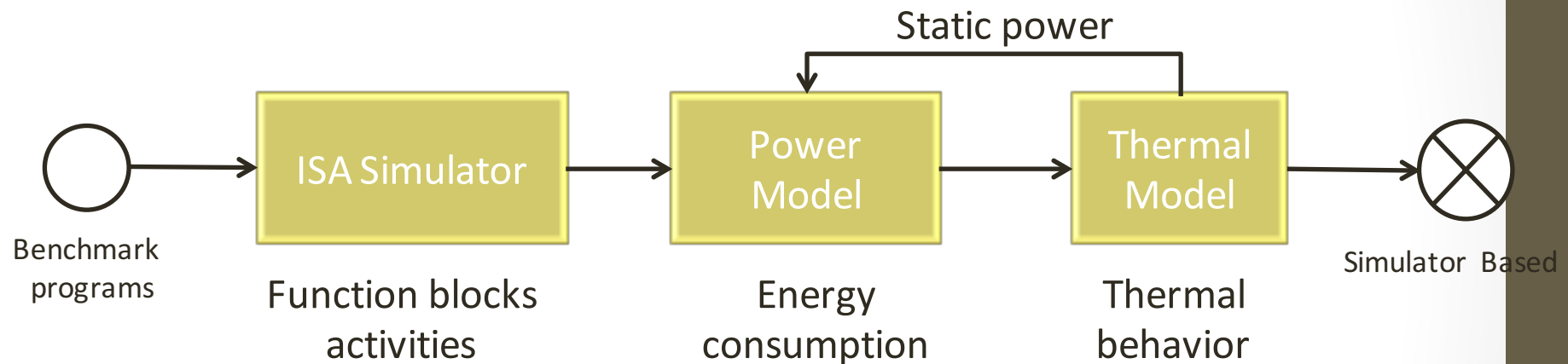
綠色運算與嵌入式系統實驗室

Leakage Power and Temperature



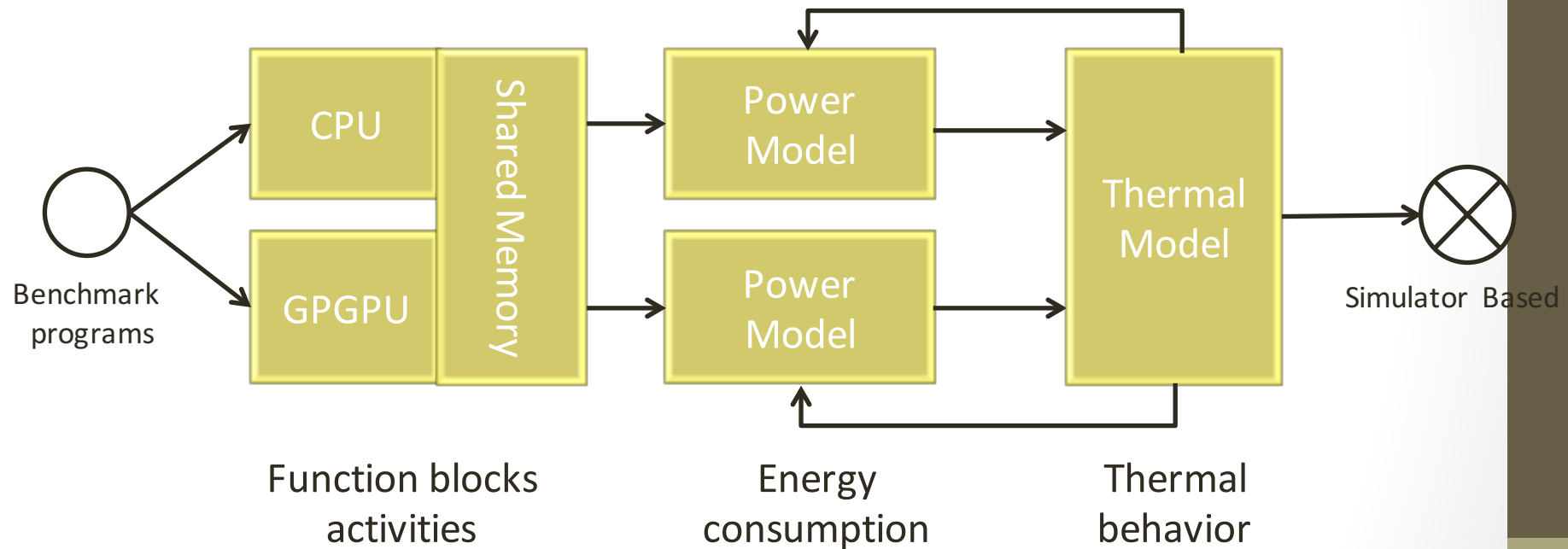
Power/Thermal Simulation

- Simulator overview



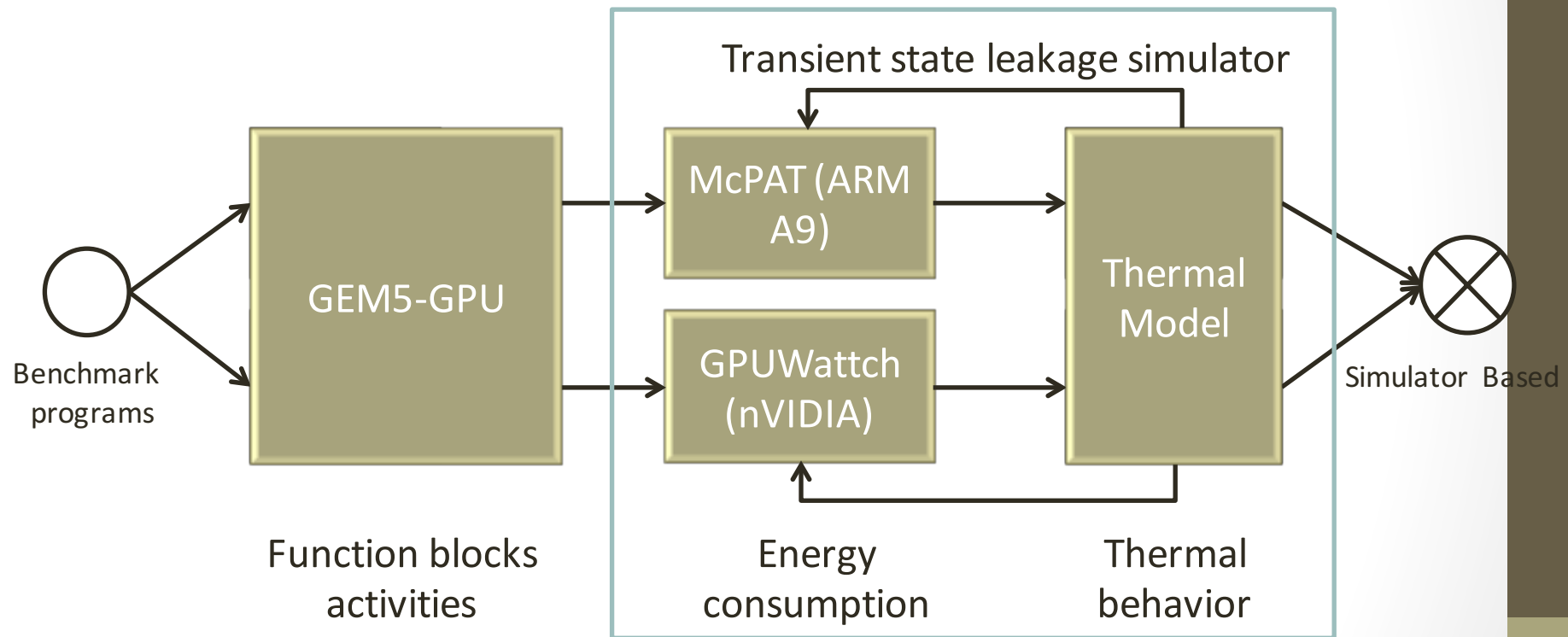
Power/Thermal Simulation

- HSA power/thermal simulator

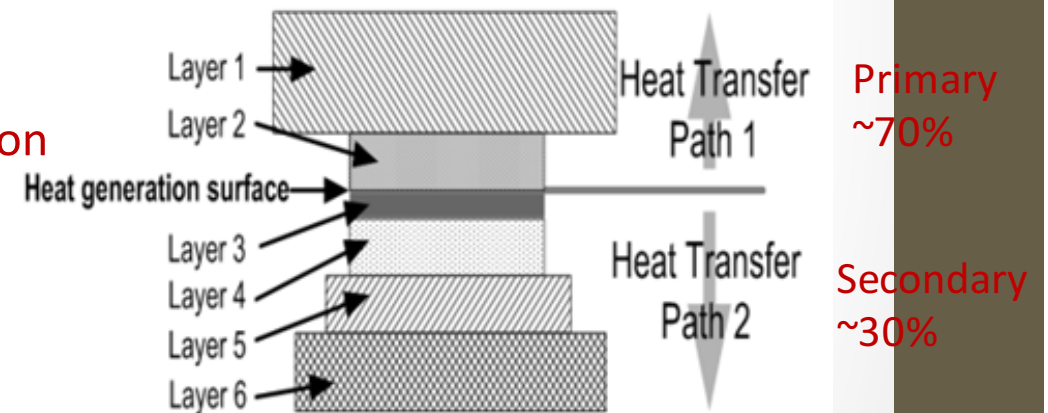
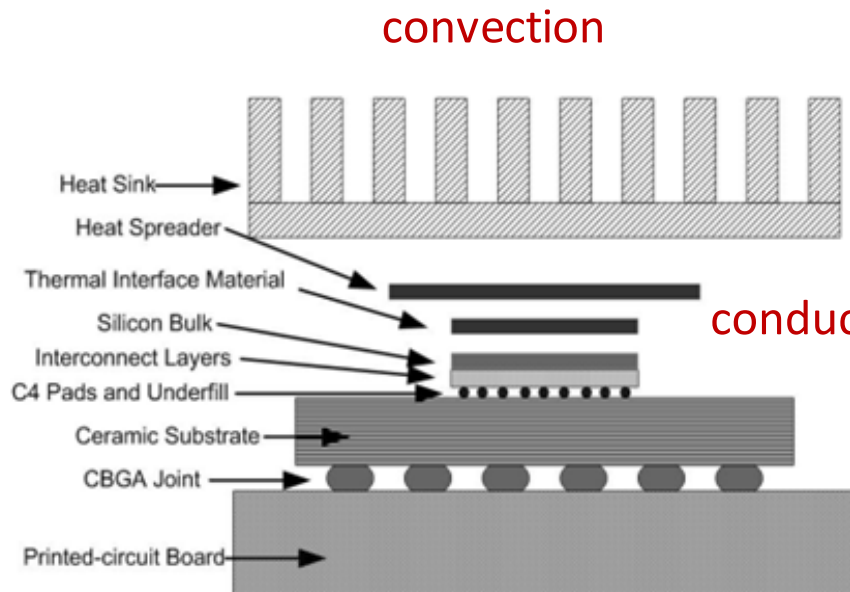
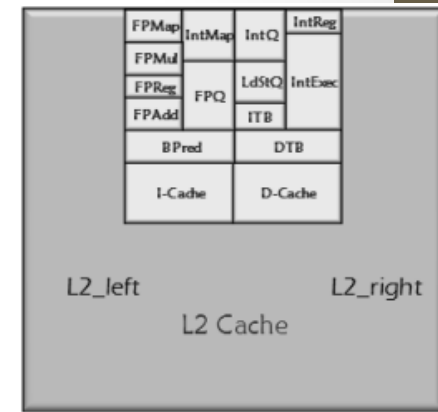


Power/Thermal Monitoring

- HSA simulator



Structure of IC



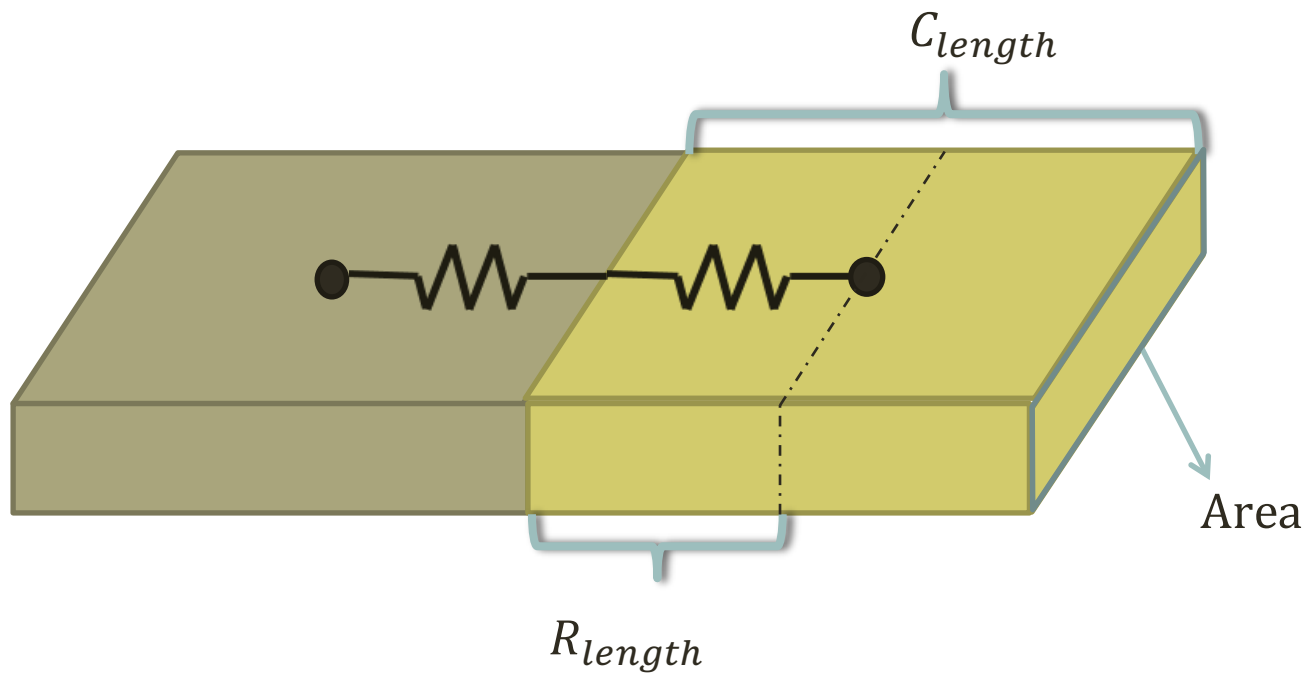
(61)

Duality between Thermal and Electrical Quantities

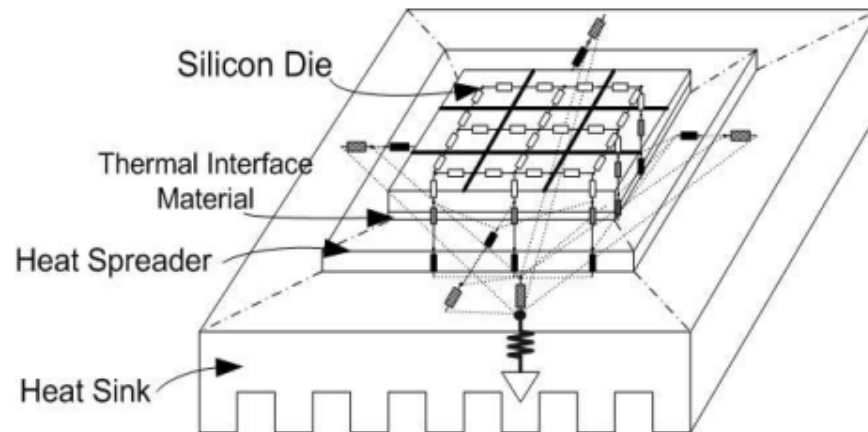
Thermal Quantity	Unit	Electrical Quantity	Unit
Heat Flow <i>input</i>	W	Current Flow	A
Temperature <i>output</i>	K	Voltage	V
Thermal Resistance	K/W	Electrical Resistance	Ω
Thermal Capacitance	J/K	Electrical Capacitance	F
Thermal RC constant	J/W	RC constant	s

Thermal Resistance and Capacitance

- $R_{green} = \frac{1}{Conductive} * \frac{R_{length}}{Area}$
- $C_{green} = C_{factor} * C_{length} * Area$

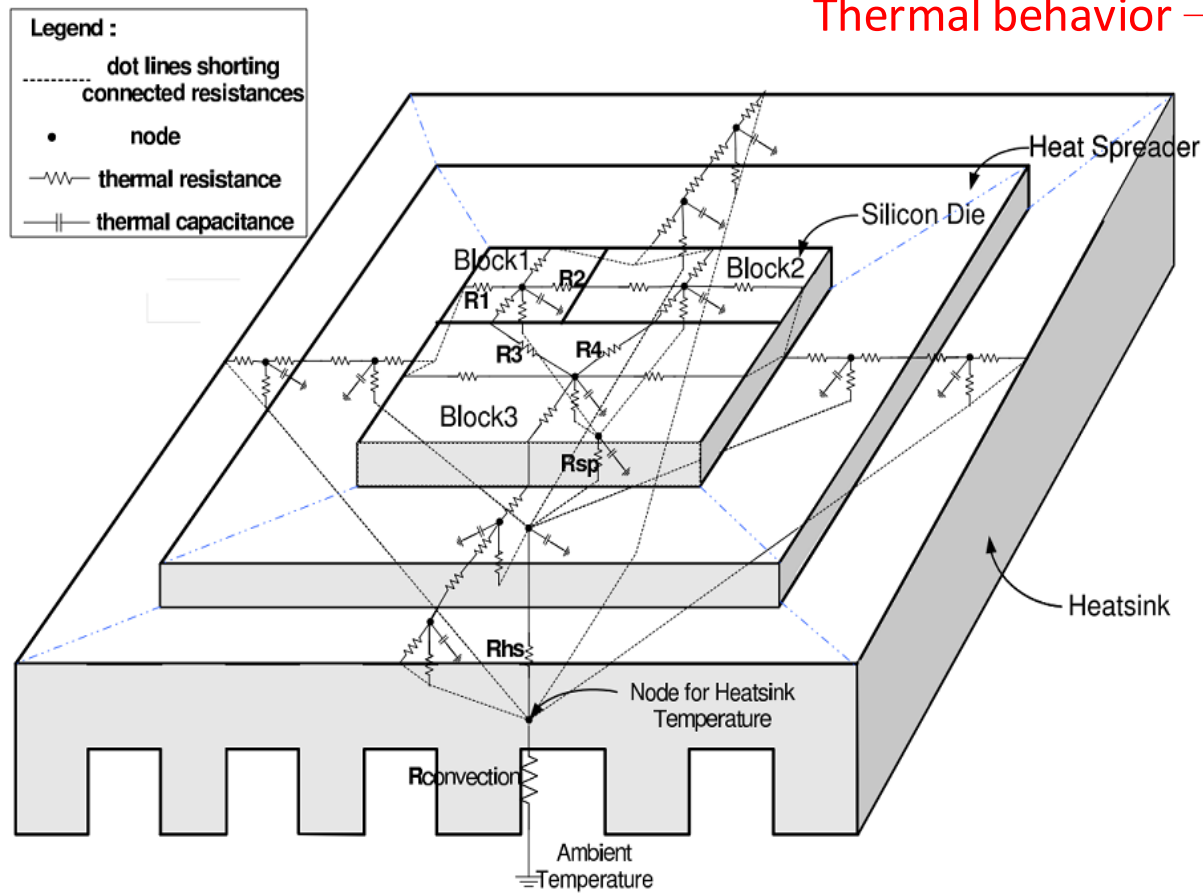


Example: Blocks with Grids



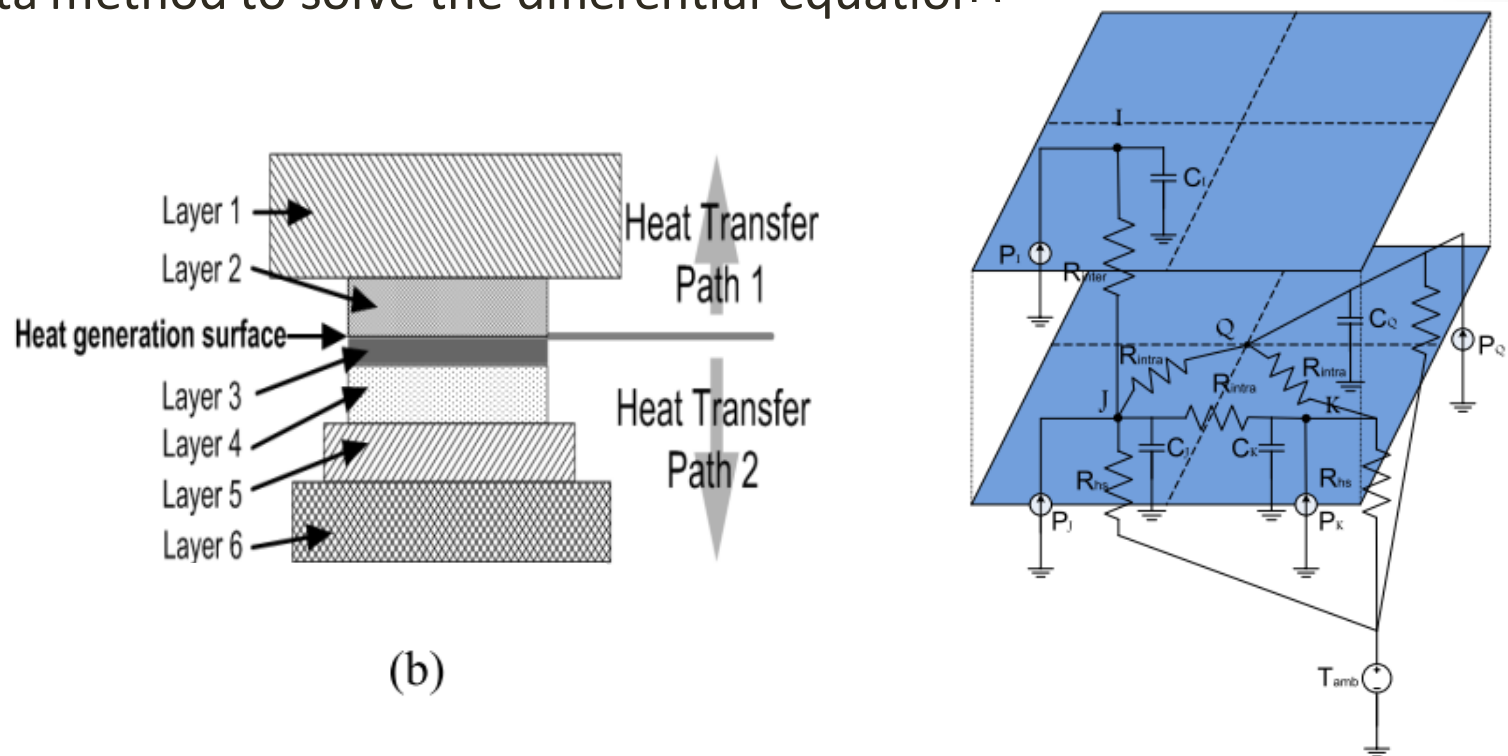
Modeling Thermal Behavior

Thermal behavior → RC model



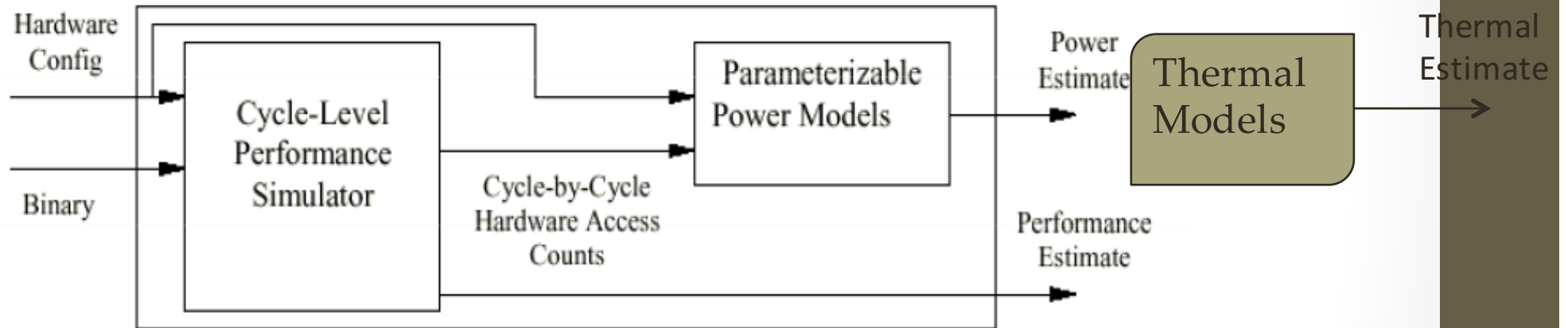
Complete Die and Package Level Thermal Model

- Derive fourth-order differential equations of the constructed dynamic compact model
- Use numerical analysis method such as the fourth-order Runge-Kutta method to solve the differential equations



Source: Wei Huang et al., "HotSpot: a compact thermal modeling methodology for early-stage VLSI design," IEEE Transactions on Very Large Scale Integration (VLSI) Systems, 2006

Thermal Simulator



Overall Structure of the Power Simulator



國立交通大學
National Chiao Tung University

Thank you